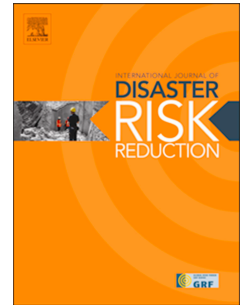


Journal Pre-proof

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Debris-Cast: A Two-Stage Machine Learning Framework for Hurricane Debris Prediction Using Multi-Source Geospatial Data

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Abstract

Accurate hurricane debris estimation is important for contractor mobilization, temporary debris-management site planning, and resource allocation. Existing tools often rely on fixed empirical formulations and limited inputs, which can restrict their ability to represent nonlinear relationships among hazard intensity, the built environment, vegetation, terrain, and flood exposure. This study evaluates a two-stage machine-learning framework for hurricane debris estimation using multi-source geospatial data, observed and post-event hazard inputs, and operational debris-removal records. The framework first classifies whether debris removal was recorded within a spatial unit and then estimates debris volume for debris-present cells. Random Forest and XGBoost models were evaluated for construction and demolition, vegetative, and combined debris from Hurricanes Helene and Milton in Pinellas County, Florida, using H3 hexagonal and equivalent-area square grids at two spatial resolutions. Classification showed higher observed performance at finer resolutions, with the highest AUC of 93.9% for construction and demolition debris using Random Forest at H3 resolution 9. Regression results revealed a scale-dependent tradeoff. Coarser grids generally produced stronger model fit and lower area-normalized error, with R^2 reaching 0.869 for vegetative debris, whereas finer grids provided greater spatial detail and lower raw per-cell MAE. Hexagonal and square grids showed comparable observed performance. The study demonstrates retrospective and screening capability using historical data. Pre-landfall forecasting, transferability across regions and events, and operational deployment require further validation using independent datasets and true forecast inputs.

Keyword: debris management, post disaster management, debris estimation, machine learning, grid geometry

Introduction

Hurricanes generate widespread physical, social, and economic impacts that often persist long after landfall. Among these impacts, disaster debris is one of the most significant challenges for post-disaster recovery because of its large volume, high removal costs, and logistical demands on local and federal agencies (Kim et al., 2018; Amini & Padgett, 2023; Cheng et al., 2024a). The financial burden can reach hundreds of millions of dollars per event (González-Dueñas et al., 2023a). For example, debris removal for Hurricane Ike (2008) accounted for nearly 34% of \$2.2 billion in Public Assistance funds, while costs reached around \$1.5 billion following Hurricane Irma (2017). Beyond cost, debris can block transportation networks, impede search and rescue operations, and delay emergency response. Floodwaters can also transport hazardous chemicals, industrial contaminants, and construction materials across affected areas, creating potential public health risks (González-Dueñas et al., 2023a; Reible et al., 2006). Accurate debris estimation is therefore essential for resource allocation, contractor mobilization, and operational planning before and after a storm.

Despite this need, commonly used debris estimation tools have important limitations. The HAZUS-MH hurricane model, one of the Federal Emergency Management Agency's (FEMA) main debris estimation platforms, runs within the ArcGIS environment and requires licensed software, specific system configurations, and trained operators, which can create barriers for some local agencies. Processing large areas at fine spatial resolution can also be slow, and the tool's structured workflow can make it difficult to connect outputs with other planning systems. Its debris estimates rely on fixed empirical formulas using a limited set of inputs. Validation studies show how these assumptions can affect model performance. When compared with landfill records following Hurricane Harvey in Beaumont, Texas, the FEMA general building formula overpredicted debris volumes by roughly an order of magnitude, and neither the FEMA nor U.S. Environmental Protection Agency's Incident Waste Decision Support Tool (I-WASTE) estimates fully captured debris generated by urban flooding (Bekkaye & Jafari, 2023). More broadly, existing debris models can carry substantial uncertainty because they rely on a small number of predictors and cannot adequately represent interactions among storm intensity, the built environment, and landscape conditions.

Machine learning offers a flexible approach to debris estimation, but, like conventional empirical models, it remains data-dependent and does not inherently guarantee greater accuracy or transferability. Its primary advantage is that it can represent nonlinear relationships among debris generation, storm hazards, the built environment, and environmental conditions without prescribing a fixed functional form. Machine learning models can also incorporate a broader range of predictors than conventional debris formulas and, once trained, rapidly recalculate debris estimates under alternative hazard inputs. Models built on publicly available geospatial data, including building inventories, flood hazard layers, vegetation indices, and meteorological products, can further reduce dependence on proprietary debris-estimation software and support integration with other planning systems. However, model transferability across regions and events,

as well as predictive uncertainty, must be evaluated separately. Important gaps nonetheless remain in how machine learning has been applied to hurricane debris estimation. Much of the existing work focuses on post-event assessment, mapping or quantifying debris after a storm has passed, rather than predicting it before landfall. Pre-landfall estimates are especially important because they allow municipalities to pre-position contractors, stage temporary debris management sites, and allocate budgets before debris removal begins. Early and accurate information can speed and reduce the costs of storm recovery.

Many high-performing approaches depend on post-event imagery, which is unavailable before landfall. Predictive geospatial features such as structural inventories, terrain, land cover, and modeled hazard fields remain underused for direct debris volume prediction. Existing debris models also rarely integrate building, environmental, and hazard features within a single framework. Moreover, the spatial units used for debris estimation are typically administrative boundaries or square grids, and the effect of spatial unit geometry on prediction performance remains largely unexamined.

This study addresses these shortcomings by developing Debris-Cast, a two-stage ML framework for hurricane debris volume prediction using multi-source geospatial features from the building, environmental, and hazard domains. The framework uses publicly available open data and is tested across four spatial grid configurations: two H3 hexagonal resolutions and two equivalent square grid resolutions. This design allows the study to evaluate how grid geometry and spatial resolution affect debris prediction performance. The framework is trained and evaluated using debris removal records from Hurricanes Helene and Milton in Pinellas County, Florida. The hazard fields used in this study are observed or post-storm reconstructed rather than forecast: storm-total rainfall, the reconstructed peak-gust wind field, and the SAR-derived flood extent all become available only after a storm has passed. The evaluation reported here is therefore a hindcast, establishing whether debris volume is predictable from multi-domain geospatial features rather than demonstrating pre-landfall forecast skill. Rainfall, wind, and track have operational forecast analogues that could be substituted, whereas observed flood extent has no pre-landfall equivalent and would need to be replaced by forecast inundation guidance or omitted. Establishing a pre-landfall capability therefore requires a separate study using true forecast products at defined lead times, with validation across additional events and regions and explicit evaluation of predictive uncertainty. The conceptual overview of the entire framework is shown in Figure 1.

This study advances hurricane debris estimation by treating debris generation as a spatially explicit, nonlinear prediction problem rather than as an output of fixed empirical coefficients. The two-stage architecture, which follows González-Dueñas et al. (2023a) in separating debris occurrence from debris-volume estimation, addresses the zero-inflated structure of debris records; the contribution of this study lies in what is combined with that structure. The tiered regression design recognizes that low-volume and high-volume debris cells may follow different predictor-response relationships. The analyses also contribute an explicit comparison of H3 hexagonal and equivalent-area square grids across two spatial resolutions, providing empirical evidence on how spatial

support and grid geometry shape both classification and regression performance. By integrating building inventories, environmental conditions, vegetation indicators, hydrometeorological hazards, evacuation zones, and flood extent into one modeling framework, the study provides a more complete representation of the interacting systems that generate hurricane debris. The results help clarify where machine learning adds value: representing nonlinear relationships across multiple predictor domains and generating spatially explicit debris estimates without prescribing fixed empirical coefficients.

Practically, the framework supports a shift from slow, formula-driven, post-hoc debris estimation toward rapid debris intelligence that can inform preparedness and early recovery. Emergency managers, public works departments, debris contractors, and state recovery agencies need spatially resolved estimates before and immediately after landfall to pre-position contractors, identify likely debris hot spots, plan temporary debris-management sites, allocate removal resources, and communicate recovery needs. Because the framework relies on open geospatial predictors and a standard spatial indexing structure, it can be integrated into operational dashboards and updated as new hazard fields become available. In its current form, the model is best positioned as a rapid post-event screening tool that can prioritize areas for field validation and resource planning. With additional validation using true forecast wind, rainfall, surge, and flood products, the model could support an anticipatory debris-estimation capability for pre-landfall planning. This practical contribution is especially important because debris removal is among the earliest recovery bottlenecks, and even approximate but spatially explicit estimates can improve readiness when the decision window is narrow.

We pursue four analytical objectives to advance rapid, spatially explicit hurricane debris estimation. First, we construct a multi-source geospatial predictor set that combines building, environmental, vegetation, and hazard variables relevant to debris generation. Second, we develop a two-stage machine-learning framework that first classifies debris occurrence and then estimates debris volume for debris-present grid cells using tiered regression, extending the two-stage structure of González-Dueñas et al. (2023a) with a magnitude-conditioned second stage. Third, we evaluate the framework across construction and demolition, vegetative, and combined debris using Random Forest and XGBoost models to assess robustness across debris streams and learning algorithms. Fourth, we compare H3 hexagonal grids with equivalent-area square grids at two spatial resolutions to determine whether grid geometry and spatial support influence debris prediction performance. These objectives are designed to address both a knowledge gap and a practice gap: the need to understand how nonlinear debris-generation processes can be modeled from open geospatial data and produce spatially consistent estimates that can support contractor mobilization, temporary debris-management site planning, resource allocation, and future pre-landfall preparedness. Success is evaluated not only by predictive metrics such as AUC, R^2 , and mean absolute error, but also by whether the framework produces interpretable, operationally usable debris estimates that can be integrated with hazard and emergency-management systems.

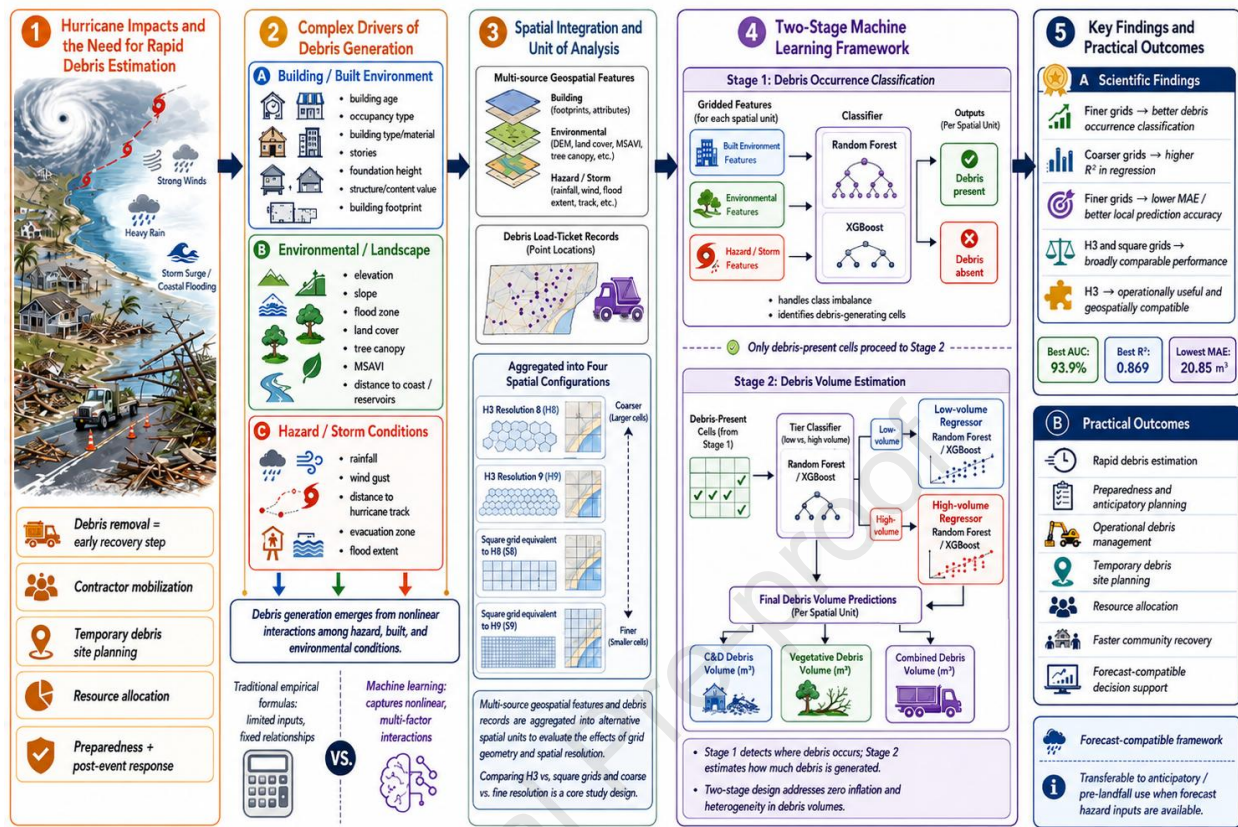


Figure 1. Conceptual overview of the study framework. The figure illustrates how hurricane debris estimation is framed as a spatial machine-learning problem driven by nonlinear interactions among hazard, building, and environmental conditions. Multi-source open geospatial data and debris records are aggregated into four spatial grid configurations (H3 and equivalent-area square grids at two resolutions) and processed through a two-stage machine-learning framework consisting of debris occurrence classification and debris volume estimation. The framework produces spatially explicit predictions of construction and demolition, vegetative, and combined debris volumes and supports rapid debris assessment and operational planning for preparedness and recovery.

Literature review

Disaster debris estimation

Accurate estimation of disaster debris is essential for emergency response, resource allocation, and recovery planning, yet the methods used to generate these estimates have changed substantially over the past decade. Early and still widely used approaches rely on empirical or physics-based formulas. The FEMA debris estimating procedure, for example, predicts single-family residential debris as a function of dwelling dimensions, the number of stories, and a vegetative cover multiplier, while the HAZUS hurricane model derives tree-debris volumes from tree density, tree height, and peak wind gusts (FEMA, 2010; Escobedo et al., 2009). The I-WASTE similarly produces order-of-magnitude estimates of debris weight and volume for disposal

planning. Although computationally simple and easy to deploy, these models were calibrated on limited historical events and generic building-stock databases, and they tend to perform poorly when transferred to new contexts.

Validation studies have repeatedly exposed the limited accuracy of these traditional tools. When compared against actual landfill records following Hurricane Harvey in Beaumont, Texas, the FEMA general building formula overpredicted debris volumes by roughly an order of magnitude and the I-WASTE estimate by approximately six times, and neither approach accounted for debris generated by urban flooding (Bekkaye & Jafari, 2023). The HAZUS tree-debris module has likewise been reported to overpredict vegetative debris by a wide margin in independent hurricane validations (Escobedo et al., 2009). More broadly, conventional debris models have been characterized as carrying errors approaching 50% because they rely on a small number of predictors and cannot represent the coupled interactions among storm intensity, the built environment, and the natural landscape (González-Dueñas et al., 2023a, 2023b). Parallel statistical efforts sought to improve vegetative-debris prediction through regression models linking urban-forest structure and storm exposure to debris volume (Escobedo et al., 2009; Thompson et al., 2011), and published literature reviews have catalogued more than twenty distinct debris-quantification methods spanning construction-and-demolition, vegetative, and household-goods debris, underscoring both the diversity of approaches and the persistent lack of a generalizable, data-driven standard (Brown et al., 2011; Marchesini et al., 2021; Kim et al., 2026).

Over the past ten years, debris research has shifted toward data-driven and machine-learning (ML) methods, propelled by the growing availability of open geospatial data, high-resolution remote sensing, and crowdsourced damage information (Al Shafian & Hu, 2024). Three broad research streams have emerged. The first predicts the quantity of debris, that is, its volume, weight, or generation rate, using supervised regression and classification on tabular predictors (e.g., González-Dueñas et al., 2023a, 2023b; Cha et al., 2020, 2022; Tajima, 2025). The second classifies or identifies debris and waste types, often from imagery, to support sorting, recycling, and disposal decisions (Zhang et al., 2023). The third maps or detects debris and related damage directly from aerial, satellite, or street-level imagery using deep learning (Cheng et al., 2024b; Yokoya et al., 2020; Yang et al., 2025). The rapid uptake of convolutional neural networks, U-Net variants, and, more recently, transformer- and foundation-model architectures for post-disaster damage assessment reflects this broader move from handcrafted formulas toward learned, multi-source representations (Al Shafian & Hu, 2024; Yang et al., 2026a, 2026b).

Within the debris-quantity stream, tree-based ensemble learners have become the dominant choice for tabular prediction. Random Forest (RF) is the most frequently adopted algorithm because it accommodates nonlinear relationships, high-dimensional and heterogeneous predictors, and mixed categorical–numerical inputs while resisting overfitting (Probst et al., 2019). RF has been used to predict building-demolition waste generation rates from structural and locational attributes, achieving coefficients of determination between roughly 0.55 and 0.80 across waste types (Cha et al., 2020), and comparative studies report that neural networks and support

vector machines can rival RF on the same task while RF remains attractive for the small, categorical-heavy datasets typical of debris research (Cha et al., 2022). For hurricane debris specifically, González-Dueñas et al. (2023a) introduced a two-stage data-driven framework that first predicts debris presence and then estimates debris amount, reporting an overall percent error of about 11%, while a companion knowledge-informed Gaussian-process model trained on 52 human-built and natural-system variables achieved normalized RMSE values near 0.49–0.50 at 250 m and 500 m resolutions (González-Dueñas et al., 2023b). The present study adopts the same two-stage logic as González-Dueñas et al. (2023a) but differs from it in predictor scope, regression design, debris categories, and the treatment of spatial unit, as detailed in the Model Framework section. Closely related damage- and loss-estimation studies adopt the same modeling philosophy: an RF classifier reached the highest accuracy (about 76%) among several algorithms for hurricane building-damage states (Klepac et al., 2022), and stacked ensembles combining RF, gradient boosting, and neural networks have been applied to hurricane-induced flood loss (Maha Arachchige & Pradhan, 2025). Beyond point predictions, Bayesian and hierarchical regression models have been used to quantify uncertainty in disaster-waste estimates, an increasingly important consideration given the scarcity of ground-truth debris records (Park et al., 2020; Tajima, 2025).

On the imagery side, deep learning has been used to segment and quantify debris from unmanned-aerial-vehicle photogrammetry (Cheng et al., 2024b), to map flood and debris-flow extent from simulation-trained U-Net architectures (Yokoya et al., 2020), and to identify disaster waste through semantic segmentation (mean IoU $\approx$ 0.80) (Zhang et al., 2023). Uncertainty-aware deep learning combined with crowdsourcing has further been proposed to translate building damage states into debris estimates (Cheng et al., 2024a).

Alongside these debris-focused efforts, post-event damage assessment from imagery has matured rapidly, which helps clarify where debris estimation fits. Large, annotated benchmarks such as xBD standardized building-damage assessment from satellite imagery and accelerated the use of deep learning for rapid post-disaster impact mapping (Gupta et al., 2019). More recent work has fine-tuned foundation vision models for post-hurricane debris segmentation, improving event-agnostic detection from aerial imagery (Amini et al., 2025), while multimodal approaches that fuse street-view imagery with structured predictors, such as wind speed, building characteristics, and distance to track, show that structured hazard and built-environment variables add predictive value beyond imagery alone (Xue et al., 2024). These advances are powerful, but they remain largely post-event and damage-oriented: they depend on imagery that is acquired, processed, and validated after impact, and they do not directly produce pre-event, spatially explicit debris-volume estimates. A parallel line of work on rapid, gridded impact nowcasting, such as machine-learning flood-damage nowcasting designed for emergency-management use (Liu et al., 2024), illustrates the value of anticipatory, spatially resolved impact intelligence built from heterogeneous predictors, the same family of decision-support tools to which rapid debris estimation belongs. This contrast frames the present study's point of departure: rather than maximizing post-event measurement accuracy from

imagery, it pursues a forecast-compatible, multi-source tabular framework that generates gridded debris occurrence and volume estimates from predictors that can be available before or shortly after landfall.

The debris targeted by these studies is heterogeneous. Hurricane and storm studies most commonly measure vegetative or urban-forest debris (Escobedo et al., 2009; Thompson et al., 2011) and construction-and-demolition or building debris (Cha et al., 2020; Cheng et al., 2024a; Mansour et al., 2026), while coastal applications additionally consider waterborne and marine debris and sediment (González-Dueñas et al., 2023a). Flood-focused work quantifies mixed flood waste and household goods (Bekkaye & Jafari, 2023; Park et al., 2020), earthquake and tsunami studies estimate structural rubble and seafloor debris (Ishibashi et al., 2021; Tabata et al., 2016). Reported units vary accordingly, including cubic yards or cubic meters of volume, tonnage or kilograms of weight, and per-unit generation rates, which complicates cross-study comparison. Notably, most hurricane debris models address a single debris type, and few jointly predict the multiple debris streams that a single storm generates (González-Dueñas et al., 2023b; Marchesini et al., 2021).

A central determinant of model performance in data-driven debris estimation is the set of predictor features, together with the modeling technique that exploits them. Across the reviewed studies, these features fall into six recurring domains: hazard and meteorological drivers, building and structural attributes, environmental and terrain characteristics, land-cover and vegetation descriptors, exposure or socioeconomic variables, and image-derived inputs. Table 1 summarizes, for each representative debris- and damage-estimation study, the modeling technique adopted, the predictor-feature domains it draws upon, its prediction setting, and its spatial unit of analysis. It shows that hazard-intensity variables (wind speed, precipitation, and flood or surge depth) and core structural attributes (building age, occupancy or use type, number of stories, and construction material) are used almost universally, whereas environmental and vegetation descriptors, such as elevation, slope, distance to water, tree canopy, and vegetation indices, appear far less consistently despite their demonstrated relevance to debris generation.

Table 1. Modeling techniques, predictor feature domains, prediction setting, and spatial unit in data-driven studies of disaster debris and damage estimation.

Study	Model / technique	Target (debris type or volume)	H	B	E	V	S	I	Prediction setting	Spatial unit / grid geometry
Escobedo et al. (2009)	Empirical / statistical	Hurricane vegetative debris volume	X			X			Pre	City / urban-forest sample plots
FEMA (2010)	Empirical formula	Residential debris volume		X		X			Pre	Jurisdiction (municipal)
Thompson et al. (2011)	Statistical regression	Hurricane urban-forest debris volume	X			X			Pre	City / urban-forest sample plots
Tabata et al. (2016)	Stock-based statistical estimation	Potential disaster waste (durables, weight)					X		Pre	Municipality
Cha et al. (2020)	RF	Demolition waste generation rate by type		X			X		Pre	Building / demolition project
Park et al. (2020)	Bayesian regression	Flood waste generation	X				X		Post	Municipality
Yokoya et al. (2020)	Attention U-Net / LinkNet (deep learning)	Flood & debris-flow mapping			X			X	Post	Image pixel
Cha et al. (2022)	Hybrid ML (ANN / RF)	Demolition waste generation rate		X			X		Pre	Building / demolition project
Klepac et al. (2022)	RF (vs. KNN, DT, GBM)	Hurricane building damage state (classification)	X	X	X				Pre	Building
González-Dueñas et al. (2023a)	ML two-stage (presence + amount)	Hurricane waterborne debris (presence & amount)	X			X			Pre	Square grid
González-Dueñas et al. (2023b)	Gaussian process (knowledge-informed)	Hurricane debris amount	X	X	X	X			Pre	Square grid (125, 250, 500 m)
Zhang et al. (2023)	DeepLabV3+ / ResNeSt (deep learning)	Disaster waste identification (segmentation)						X	Post	Image pixel
Cheng et al. (2024a)	Uncertainty-aware deep learning + crowdsourcing	Post-hurricane building debris						X	Post	Building
Cheng et al. (2024b)	Region-based CNN + photogrammetry	Debris detection & volume (UAV)						X	Post	Detected object (UAV image)
Maha Arachchige & Pradhan (2025)	Stacked ensemble (RF + XGBoost + GBM + NN)	Hurricane flood loss / damage	X	X				X	Post	Not reported
Tajima (2025)	Bayesian hierarchical regression	Disaster waste amount (with uncertainty)					X		Pre	Municipality

Note. Predictor feature domains: H = hazard / meteorological (e.g., wind speed, precipitation, storm surge or flood depth, distance to storm track); B = building / structural (e.g., age, occupancy or use type, number of stories, construction material, floor area, value); E = environmental / terrain (e.g., elevation, slope, distance to water, catchment and channel characteristics); V = land cover and vegetation (e.g., land use, tree canopy, NDVI / MSAVI); S = exposure and socioeconomic (e.g., population or damaged houses, urbanization, evacuation zone); I = image-derived inputs (e.g., aerial / UAV, satellite, SAR / DEM, or video). An X indicates that the domain was used. González-Dueñas et al. (2023a, 2023b) report features at the category level. Prediction setting: Pre = the model can be executed with inputs

10 *available before impact; Post = the model requires imagery or damage observations collected after the event. Spatial unit is the unit over which predictors and debris*
11 *targets are aggregated, with grid resolution given where the study specifies it. "Not reported" indicates an attribute that the source does not report at a comparable*
12 *level.*

Two further attributes of these studies are reported in the last two columns of Table 1 and are elaborated here because they are central to the gaps identified below: the prediction setting, defined by whether a model can be executed with inputs available before impact or instead requires observations collected afterwards, and the spatial unit over which predictors and debris targets are aggregated.

The reviewed work divides on prediction setting according to what each model requires at inference time rather than according to modeling sophistication. Empirical formulas and statistical regressions (FEMA, 2010; Escobedo et al., 2009; Thompson et al., 2011), stock-based waste accounting (Tabata et al., 2016; Tajima, 2025), building-level waste and damage models (Cha et al., 2020, 2022; Klepac et al., 2022), and the data-driven hurricane debris models of González-Dueñas et al. (2023a, 2023b) can in principle all be run before an event, because their predictors are structural, terrain, land-cover, or hazard fields for which forecast equivalents exist. By contrast, the deep-learning studies that report the strongest performance (Yokoya et al., 2020; Zhang et al., 2023; Cheng et al., 2024a, 2024b) and the waste and loss models that key on observed damage (Park et al., 2020) require post-event imagery or damage records and therefore cannot inform pre-positioning decisions.

On spatial unit, every reviewed study inherits its unit of analysis from its data source rather than selecting it as a design variable: individual buildings or demolition projects (Cha et al., 2020, 2022; Klepac et al., 2022), municipal or jurisdictional boundaries (FEMA, 2010; Tabata et al., 2016; Park et al., 2020; Tajima, 2025), cities and urban-forest sample plots (Escobedo et al., 2009; Thompson et al., 2011), image pixels or detected objects (Yokoya et al., 2020; Zhang et al., 2023; Cheng et al., 2024a, 2024b), and square grids at fixed resolutions in the only gridded hurricane debris studies (González-Dueñas et al., 2023a, 2023b). No reviewed study treats the spatial unit as an experimental factor, and none applies hexagonal indexing.

This synthesis exposes several gaps in current data-driven disaster debris estimation research.

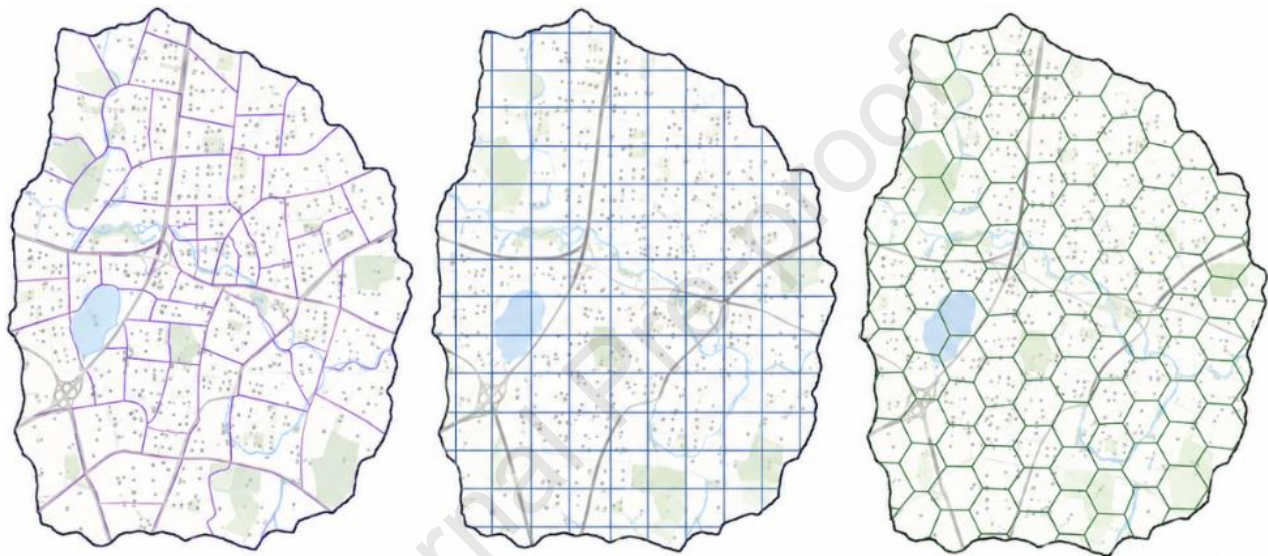
First, pre-event capability and modeling expressiveness are misaligned: the frameworks that can be run before landfall are the least expressive, and the most expressive are tied to post-event imagery, even though anticipating debris volume before impact is what enables municipalities to pre-position contractors, stage temporary debris-management sites, and allocate budgets (González-Dueñas et al., 2023a). Second, predictive geospatial features, such as structural inventories, terrain, land cover, and modeled hazard fields, remain comparatively underexploited for direct volume prediction. Third, existing debris models rarely integrate building, environmental, and hazard features within a single framework, and the spatial unit over which they operate remains an inherited property of the data rather than a design choice, so its influence on debris prediction is untested (González-Dueñas et al., 2023b). Fourth, several frequently relevant predictors, including tree canopy, vegetation indices, evacuation zoning, and fine-scale flood extent, are seldom included together, as Table 1 makes clear. Taken together, these gaps point to clear directions for future work: predicting hurricane debris volume before landfall from multi-source, pre-disaster

geospatial features that jointly span the building, environmental, and hazard domains, and examining how the choice of spatial unit, including hexagonal (H3) and square grid geometries, shapes such predictions. In this study, the hazard fields are observed or post-storm reconstructed conditions for Hurricanes Helene and Milton in 2024, so the evaluation reported below is a hindcast rather than a forecast. Whether the framework transfers to a genuine pre-landfall setting depends on substituting forecast products at defined lead times, which lies outside the scope of this study.

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Spatial unit and grid geometry in geospatial data analytics

The choice of spatial unit shapes how hazard processes are represented, how features are aggregated, and ultimately how well a model can learn from the data. In disaster research, three spatial frameworks dominate: administrative boundaries (Bekkaye & Jafari, 2023; Mansour et al., 2026), square grids (Amini & Padgett, 2023; González-Dueñas et al., 2023a; González-Dueñas et al., 2023b), and hexagonal grids (Birch et al., 2007; Bousquin, 2022) (see Figure 2). Each carries distinct strengths and limitations that become particularly consequential when modeling spatially continuous processes like hurricane debris generation.



(a) Administrative boundary

(b) Square grid

(c) Hex Grid

Figure 2. Comparison of spatial unit frameworks. The panels show (a) administrative boundaries, (b) regular square grids, and (c) hexagonal (H3) grids, illustrating how the choice of spatial unit governs the aggregation of hazard and built-environment features and the representation of debris-generation processes.

Administrative boundaries such as census blocks, tracts, and counties are the most common unit of analysis in debris estimation research (Li et al., 2025). Their appeal is practical: they align with how agencies collect, report, and act on data, making results directly usable for governance and resource allocation. However, administrative boundaries are defined by political and historical factors, not by the physical processes being studied. This mismatch introduces aggregation bias and gives rise to the Modifiable Areal Unit Problem (MAUP), where statistical relationships shift depending on the scale and shape of the spatial units imposed on the analysis (Openshaw, 1984). Under MAUP, patterns that exist at fine scales can be masked or distorted at coarser administrative levels, reducing the reliability of model outputs and complicating cross-event comparisons (Hayward, 2009; Pollack et al., 2022).

Square grids offer a more consistent alternative. Their regular structure aligns naturally with raster data formats, which are standard for hazard products like wind fields, storm surge models, and flood extent layers, making feature extraction straightforward (Wang & Ai, 2018). Compared to administrative boundaries, square grids reduce aggregation bias and better preserve local spatial variability, particularly in urban environments where hazard exposure and building characteristics can change substantially over short distances (Chinthavali et al., 2024; Li et al., 2025). Most flood and urban disaster studies use square grid cells ranging from 30 meters to 1 km. For debris estimation specifically, González-Dueñas et al. (2023b) used square grids at 125-, 250-, and 500-meter resolutions and found that model performance improved at coarser resolutions, consistent with how spatial aggregation affects regression noise. Despite these advantages, square grids introduce directional bias because diagonal neighbors are farther apart than orthogonal ones. This unequal neighbor distance can distort analyses involving neighborhood interactions, connectivity, or the propagation of hazard impacts across space, all of which are relevant to debris generation modeling (see Figure 3).

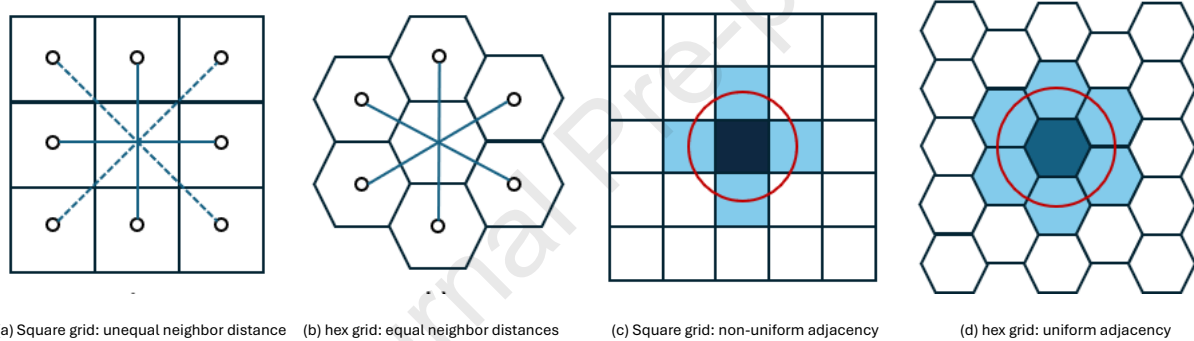


Figure 3. Comparison of neighbor distance and spatial adjacency between square and hexagonal grid tessellations. The panels contrast the unequal orthogonal-versus-diagonal neighbor distances and non-uniform adjacency of square grids with the uniform six-neighbor adjacency of hexagonal grids, highlighting the directional bias that grid geometry can introduce into spatial analysis.

Hexagonal grids address this limitation. Each hexagonal cell shares an equal-length edge with all six neighbors, ensuring uniform adjacency distances and removing the directional bias inherent to square tessellations (Chinthavali et al., 2024; D'Ambrosio et al., 2003; Wang & Ai, 2018; Wang & Kwan, 2018). Hexagons also have a lower perimeter-to-area ratio than squares or triangles, which reduces edge effects and produces more compact, spatially coherent cells (Lin & Peng, 2001; Wang & Ai, 2018). In Figure 3, panels (a) and (c) show square grids, while panels (b) and (d) show hexagonal grids. In square grids, diagonal neighbors lie farther away than orthogonal neighbors, introducing directional bias (a), and the inscribed circle extends beyond adjacent cells, reflecting unequal adjacency distances (c). Hexagonal grids maintain equal distances to all six neighbors (b), and the inscribed circle aligns with the immediate neighborhood, reflecting uniform spatial adjacency (d). These properties make hexagonal grids better suited to representing curvilinear and radial spatial patterns, such as storm-surge propagation, wind swaths, and flood extent, which do not follow orthogonal paths (Birch et al., 2007; Bousquin, 2022). Hexagonal tessellations have been

applied across ecology, hydrology, cartography, and communications, and have been adopted for flood-risk mapping and coastal hazard characterization. The Hexagonal Hierarchical Geospatial Indexing System (H3), developed by Uber Technologies, has become the most widely used standardized hexagonal grid system in applied geospatial work, offering 16 hierarchical resolution levels and broad compatibility with modern geospatial platforms and hazard-modeling tools. Svelling et al. (2026) applied H3 grids at resolutions 6 through 13 for pluvial flood-risk assessment and found that spatial patterns were preserved across resolutions, although finer grids captured localized extremes that coarser aggregations missed. Li et al. (2022) similarly used hexagonal discrete global grids to model future flood risk at multiple scales, demonstrating the value of hierarchical hex systems for multi-resolution disaster analysis.

Despite this growing adoption in hazard modeling, hexagonal grids have not been applied to debris estimation. Existing debris models rely on administrative units or square grids (Bekkaye & Jafari, 2023; González-Dueñas et al. 2023b), and no study has examined whether grid geometry shapes debris prediction outcomes. This is a meaningful gap. If hexagonal grids perform comparably to square grids for debris estimation, then the operational benefits of H3, including its standardized scale, native compatibility with other geospatial hazard layers, and ease of spatial integration, provide a practical case for adopting it (Chinthavali et al., 2024) as a default framework for debris planning. This study tests that question by comparing H3 hexagonal and square grid configurations across two spatial resolutions, using the same ML framework and feature set for both.

Despite substantial progress in disaster debris modeling, three gaps continue to limit the scientific and operational value of existing approaches. First, conventional tools and empirical formulas remain constrained by small sets of predictors and fixed assumptions, making them poorly suited to represent the nonlinear interactions among hazard intensity, the built environment, vegetation, terrain, flood exposure, and cleanup logistics that shape debris generation. Second, much of the recent data-driven debris literature remains either post-event in orientation or dependent on imagery and records that are not available before landfall, limiting its usefulness for anticipatory contractor mobilization, temporary debris-management site planning, and pre-event budgeting. Third, prior studies have rarely examined how the spatial unit of analysis affects debris estimation, particularly whether H3 hexagonal grids can provide operational integration advantages without sacrificing predictive performance relative to conventional square grids. This study departs from prior work by developing a two-stage machine-learning framework that combines open building, environmental, vegetation, and hazard predictors with operational debris-removal records to estimate debris occurrence and debris volume across debris types, algorithms, grid resolutions, and grid geometries. The point of departure is neither the use of machine learning nor the two-stage structure itself, which follows González-Dueñas et al. (2023a); it is the integration of nonlinear debris modeling, multi-source geospatial predictors, and spatial-index design into a framework that can support rapid debris estimation and future pre-landfall decision support.

The methodological gap in hurricane debris estimation stems not from a lack of computational sophistication in existing models; rather many approaches remain misaligned with the structure of

the operational problem. Formula-based tools such as HAZUS-MH, FEMA debris-estimation procedures, and related planning tools are useful for standardized screening, but they rely on a limited set of inputs and are not designed to learn nonlinear interactions among hazards, structures, vegetation, terrain, and flood processes. Post-event remote-sensing and image-based methods can provide valuable damage and debris evidence after impact, but they cannot fully support decisions that must be made before imagery is acquired, processed, and validated. Prior machine-learning studies have improved debris estimation, yet most have not simultaneously addressed the zero-heavy nature of debris observations, multiple debris streams, multi-domain open geospatial predictors, spatial-resolution sensitivity, and the operational implications of grid geometry. These gaps limit both scientific interpretation and practical adoption.

This paper's point of departure is the implementation of a two-stage spatial learning framework designed around the real structure of debris operations. The first stage estimates whether a grid cell is likely to generate debris, directly addressing the prevalence of zero-debris cells; the second stage estimates volume only for debris-present cells, allowing the model to focus on magnitude after occurrence has been established. The framework further distinguishes low-volume and high-volume debris through tiered regression, recognizing that the determinants of minor localized debris may differ from those associated with large debris accumulations. By applying the same modeling architecture to construction and demolition, vegetative, and combined debris across Random Forest and XGBoost models, the study evaluates whether the approach is robust across debris streams and algorithms. Finally, by comparing H3 hexagonal grids with equivalent-area square grids, the paper moves beyond model selection alone and examines the spatial design of debris intelligence, asking whether a standardized hexagonal index can support operational integration while maintaining predictive performance. In this study, H3 and equivalent-area square grids were compared based on model performance after aggregating the same predictor and debris datasets. Because the Random Forest and XGBoost models treated each cell independently, the study did not evaluate neighborhood connectivity, directional bias, or hazard propagation. These potential advantages would require a spatial model that explicitly uses adjacency relationships.

Methods

Study area

Pinellas County is a coastal county in west-central Florida, United States, occupying a narrow peninsula bounded by Tampa Bay to the east and the Gulf of Mexico to the west. With a population of approximately 960,000 residents and an area of roughly 1570 sq. km (608 sq. miles), it ranks as the seventh most densely populated county in Florida and contains a mix of residential, commercial, and industrial land uses across 24 municipalities (U.S. Census Bureau, n.d.). The county's low-lying coastal geography (mean elevation of 4 meters above sea level), extensive shorelines, and high population density make it particularly susceptible to hurricane-induced

hazards, including storm-surge inundation, high winds, and heavy rainfall, marking it as a region of high disaster risk.

We focus on the impacts of two consecutive major hurricanes that made landfall in rapid succession during the 2024 Atlantic hurricane season: Hurricane Helene and Hurricane Milton. Hurricane Helene made landfall in the Florida Bend region on 27 September 2024 as a Category 4 hurricane, producing catastrophic storm surge, widespread flooding, and extensive debris generation (Hagen et al., 2026). Shortly thereafter, Hurricane Milton made landfall on Siesta Key, Florida, on 10 October 2024 as a Category 3 hurricane, subjecting the region to a second episode of hurricane-induced hazards before communities had fully recovered from Hurricane Helene (Beven et al., 2025). The sequential occurrence of these two major hurricanes within a short timeframe presented an unprecedented dual-hazard exposure scenario, resulting in compounded debris accumulation across the county and posing significant challenges to debris-removal operations. The geographic location of Pinellas County, together with the tracks of Hurricane Helene and Hurricane Milton, is illustrated in Figure 4.

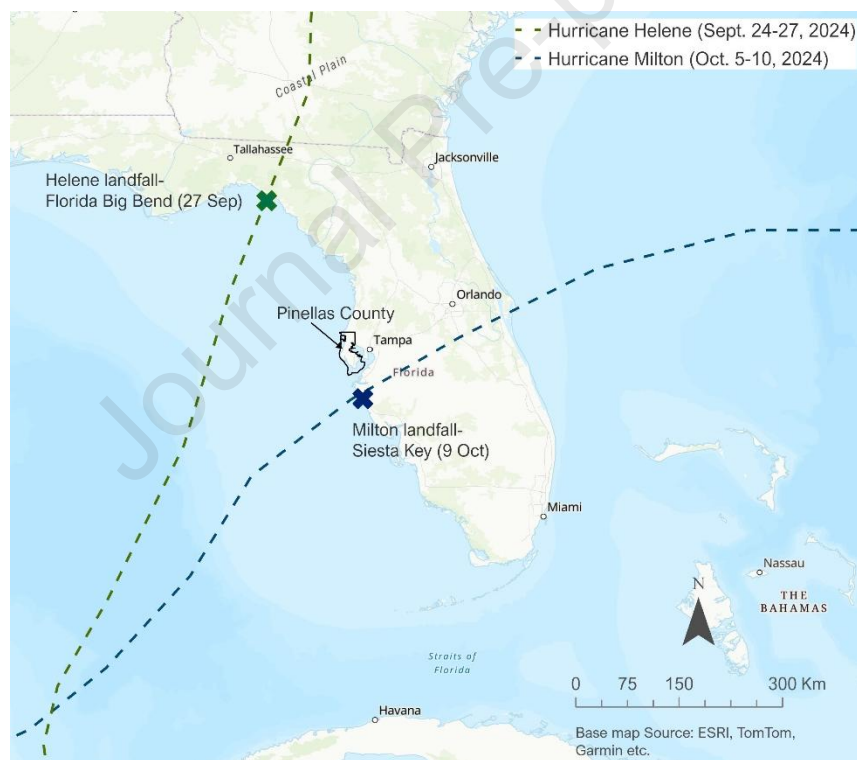


Figure 4. Tracks of Hurricane Helene and Hurricane Milton across the Gulf of Mexico and Florida, showing the two sequential 2024 landfalls that produced the compound debris-generating event analyzed in Pinellas County. Note: track data source: NOAA National Hurricane Center Best Track GIS Archive. Landfall locations verified using NHC Tropical Cyclone Reports..

Data description

The debris data were aggregated from two publicly accessible ArcGIS Online datasets covering Pinellas County and the City of St. Petersburg, Florida (see Table 2). Each dataset consists of

operational load-ticket data generated during the debris-cleanup effort following Hurricanes Helene and Milton. Each record represents a debris-removal event and includes information such as a load-ticket ID, GPS coordinates, debris type, and debris volume.

The response variable is the debris-removal volume recorded on the operational load tickets and aggregated to spatial units using the load GPS coordinates. This volume does not directly measure the amount of debris physically generated at those coordinates. Debris may be generated at damaged structures or vegetated areas, moved and accumulated at curbs, properties, parks, or citizen drop-off locations, and then collected and transported through the contractor-managed debris-removal system. The recorded coordinates therefore represent loading or collection locations, which may differ from the original debris-generation locations. Debris managed outside the recorded collection system is not represented in the datasets. Because cleanup activities associated with Helene and Milton overlapped, the records represent cumulative removal activity from the sequential events and cannot be assigned reliably to an individual hurricane.

The datasets were downloaded from their ArcGIS Feature Servers using a Python script. After download, an initial triage assessed dataset composition, field completeness, debris-type coverage, and debris-volume ranges. The Pinellas County dataset contained 18,278 records, and the St. Petersburg dataset contained 71,766 records. Across both datasets, every sample contained a load-ticket ID, load date, load GPS coordinates, and volume. All records in the Pinellas County dataset included a debris type, whereas only 54.2% of St. Petersburg records did (see Table 3).

Table 2. Debris dataset summary

Category	Pinellas County	City of St. Petersburg	Combined
Records	18,278	71,766	90,044
Columns	54	55	-
Time range	Oct. 2, 2024 – Apr. 10, 2025	Oct. 3, 2024 – Feb. 1, 2025	Oct. 2, 2024 – Apr. 10, 2025
Total debris volume	493,101.68 m ³	2,721,300.67 m ³	3,214,402.35 m ³
Mean volume per record	26.98 m ³	37.92 m ³	-
Median volume per record	26.76 m ³	31.65 m ³	-
Load date completeness	100%	100%	100%
GPS coordinate completeness	100%	100%	100%
Volume completeness	100%	100%	100%
Debris type completeness	100%	54.2%	~63.5%

Table 3. Disaster debris dataset breakdown by debris type

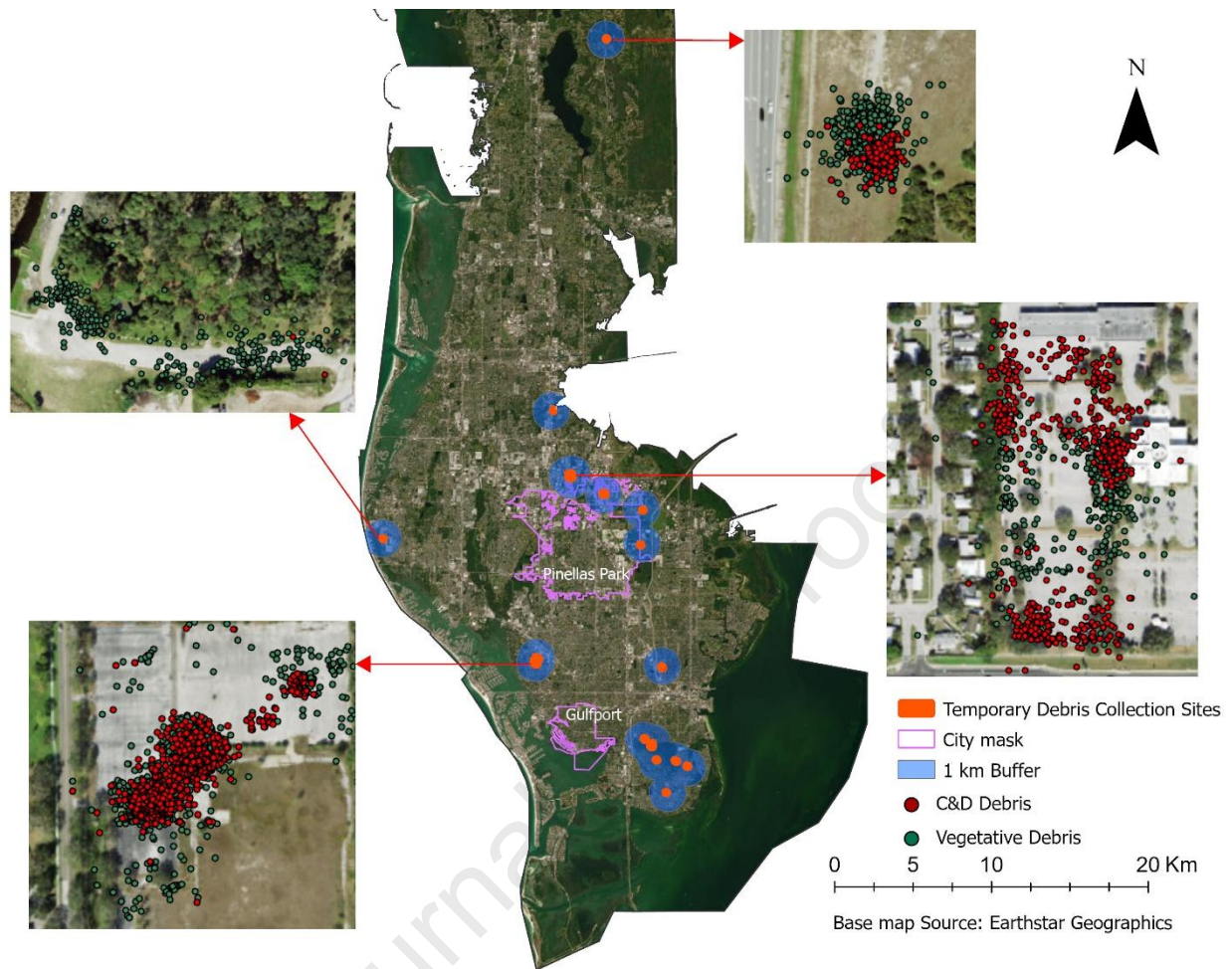
Debris Type	Pinellas records	Pinellas volume	St. Petersburg records	St. Petersburg volume
vegetative/woody	13,761	364,110.92 m ³	24,237	1,020,937.66 m ³

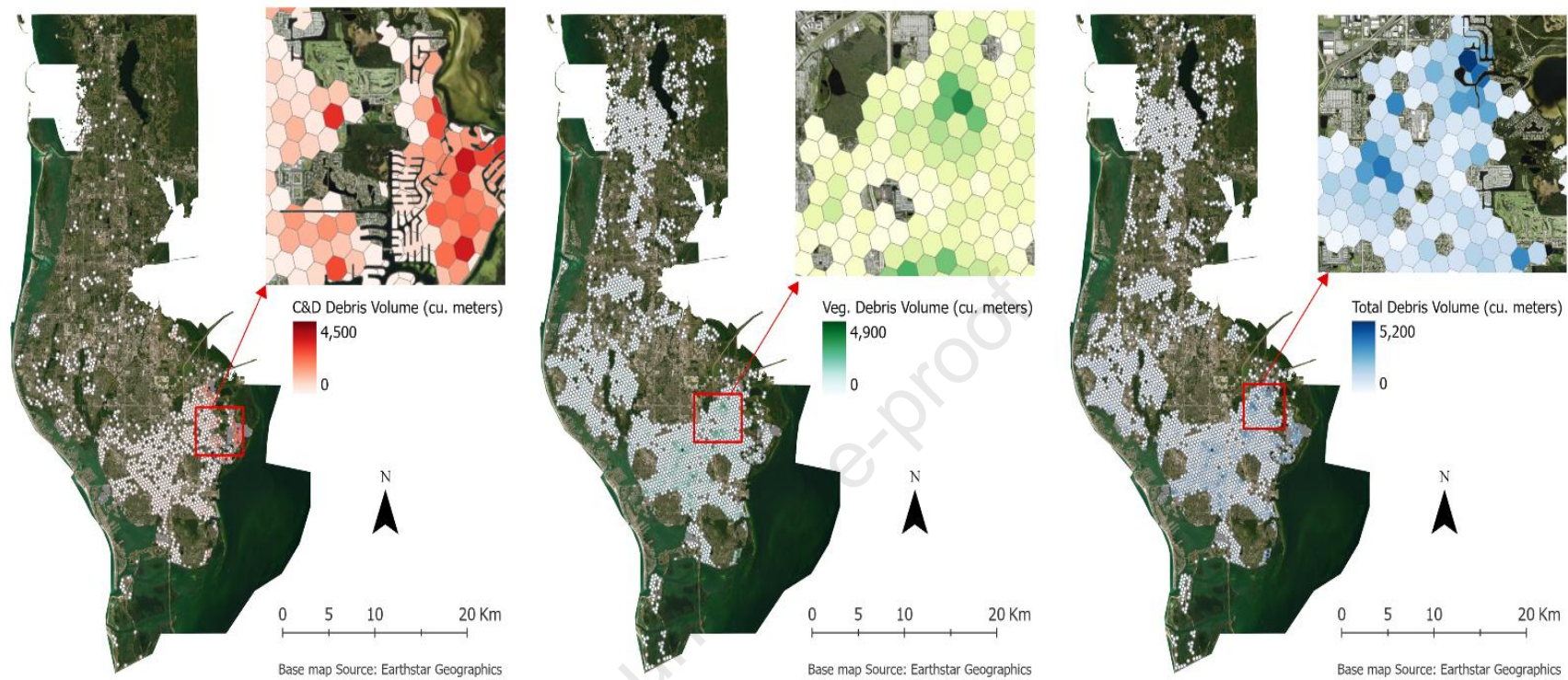
construction & demolition	2,490	67,951.27 m ³	7,958	324,044.38 m ³
citizen dropoff debris - vegetative/woody	-	-	4,085	184,752.50 m ³
citizen dropoff debris - c&d	-	-	2,632	122,836.02 m ³
veg park debris	1,843	54,713.34 m ³	-	-
Other	184	6,326.15 m ³	-	-
Missing/blank	0	0	32,854	1,068,730.19 m ³

Post Data Processing

The two datasets were merged and examined for spatial overlap to eliminate duplicate collection points. The analysis included debris removal tickets from both Hurricane Helene and Hurricane Milton. Two key attributes were retained: debris type and debris volume. Debris types were reclassified into two categories. Entries such as 'Construction & Demolition (C&D)', 'Private Property Debris Removal (PPDR)/Commercial Property Debris Removal (CPDR) C&D', 'C&D park debris', and 'citizen drop-off debris C&D' were grouped as Construction and Demolition (C&D) debris, while 'vegetative/woody', 'vegetative park debris', 'PPDR/CPDR vegetative', and 'citizen drop-off debris vegetative' were categorized as Vegetative debris. Records without a specified debris type were excluded to retain only complete and verifiable observations. This ensured that debris volumes could be reliably assigned to the correct debris category and spatial location without introducing assumptions or potential misclassification. As a result, the response variable used for model training and validation remained consistent and well defined. Debris volume represents the amount collected by contractors and is measured in cubic meters. The final analysis was conducted using three datasets: C&D debris, vegetative debris, and combined debris.

Clusters of closely located collection points were identified as likely temporary debris collection sites and removed as shown in Figure 5. Because debris may have been recorded both at individual collection points and again after being aggregated at temporary debris sites, these locations could introduce potential double counting. To reduce this risk, a 1 km buffer was generated around the identified temporary sites, and grids within these buffers were excluded from the analysis on the assumption that debris from surrounding neighborhoods had been transported and aggregated at these locations. In addition, because debris collection data were not available for all municipalities within Pinellas County, cities with incomplete records, including Pinellas Park and Gulfport, were excluded from the analysis. The grids were further removed by masking waterbodies present in the study area. Figure 6 illustrates the distribution of the debris volumes by debris type in Pinellas County.





(a) Construction & Demolition debris

(b) Vegetative debris

(c) Combined debris

Figure 6. Spatial distribution of collected debris volume across Pinellas County, Florida, shown separately for different debris types, illustrating the spatial concentration and heterogeneity of debris generation across the county.

Spatial Feature processing and Grid-Level Aggregation

The model features were derived from multiple data sources and organized into three domains: building, environmental, and hazard.

A. Building features

Building related data were obtained from the 2022 National Structure Inventory (NSI) developed by the U.S. Army Corps of Engineers (2022). The NSI provides point based structural information with attributes relevant to assessing risks from natural and human induced hazards. From the available attributes, eight variables relevant to debris generation during hurricane events were selected. Preprocessing was performed using the Spatial Join tool in ArcGIS Pro 3.5.3 to aggregate attributes to the different grid resolutions and geometries. Building age was estimated by subtracting the construction year from 2026, and mean values were computed for each grid. Occupancy and building types were aggregated as counts within each grid. Occupancy types included residential, industrial, commercial, and other categories such as government and educational uses, while building types included steel, masonry, wood, and concrete structures. Total building acreage was computed as a sum, whereas mean values were assigned to each grid for foundation height, number of stories, and structural and content values.

B. Environmental features

Five environmental variables were incorporated. Distances from each spatial grid to nearby water bodies, including coastlines and reservoirs, were calculated using the Near tool in ArcGIS Pro. Proximity to these features is expected to increase exposure to storm surge and flooding, thereby influencing debris generation. Elevation and slope were derived from NASA digital elevation data, and average values were assigned to each grid. Although slope and elevation do not directly generate debris, they influence the spatial distribution of runoff, surge, and wave energy, thereby affecting debris accumulation patterns. Flood hazard zones were obtained from FEMA's National Flood Hazard Layer (NFHL) and assigned to grids using a spatial join tool with the largest overlap criterion. Land cover data from the Florida Department of Environmental Protection (FDEP) were processed similarly to extract categorical attributes. Vegetation related variables included tree canopy cover was obtained from the USDA Forest Service for year 2023 and was quantified as the mean percentage cover within each grid. A second vegetation feature, the Modified Soil-Adjusted Vegetation Index (MSAVI), characterized the spatial distribution and density of vegetation across Pinellas County. Using Landsat imagery of 2024, the MSAVI map was generated in Google Earth Engine, and the percentage cover was assigned to each unit of analysis.

C. Hazard features

The hydrometeorological conditions associated with Hurricane Helene and Hurricane Milton served as the primary hazard-related drivers of debris generation. Precipitation was derived from the NASA IMERG dataset (NASA GPM, n.d.) for 25–28 September 2024 (Hurricane Helene) and 5–10 October 2024 (Hurricane Milton). Wind intensity data were obtained from the National Institute of

Standards and Technology (NIST) in collaboration with Applied Research Associates (ARA) and represent three-second peak gusts at 10 m height over open terrain, consistent with modern engineering standards (Levitan et al., 2024). Average wind intensity values were assigned to each grid. Distance to the hurricane track, obtained from the National Hurricane Center (NOAA NHC, n.d.), was calculated using the Near tool. Evacuation data from the Florida Division of Emergency Management (FDEM) were used to assign categorical evacuation levels to each grid based on the largest spatial overlap. Finally, flood extent data from the NASA Disasters Mapping Portal were used to generate a binary flood-presence map and calculate the percentage of flooded areas within each grid. The products were developed by the Alaska Satellite Facility using Sentinel-1 SAR imagery and radiometrically terrain-corrected RGB composites, where blue generally represents smooth surfaces such as water present at the time of the satellite overpass. For Helene, pre-event imagery was acquired on September 14, 16, 19, and 21, 2024, and post-event imagery on September 26 and 28 and October 3, 2024. For Milton, pre-event imagery was acquired on October 3 and 8, 2024, and post-event imagery on October 11 and 20, 2024.

The summary of the data and their sources are shown in Table 4:

Dataset	Features	Description	Update frequency	Data version/Acquisition dates	Resolution	Source
Building	Year built	Describes the median year built of structures within the Census Block.	Data updated in 2011, 2015, 2019, 2022	2022	Point data	NSI USACE
	Occupation type	Based on FEMA occupancy type definitions; help determine structure valuation and define structure damage				
	Building type	Structure type associated with exterior wall and used for structure stability functions				
	Acreage	Estimated square footage of the structure				
	Foundation height	Describes the foundation height of the structure in feet from the ground elevation.				
	Number of stories	The number of stories of the structure.				
	Structure values	Value in dollars of the structure.				
	Content values	Value in dollars of the contents of the structure.				
Environmental	Coastlines	Coastal distance to each grid.	Annual; last updated 2025	2024	Polyline data	US Census TIGER
	Slope	The average measure of the incline or steepness of the land surface.	Acquired year - 2000	2000	~ 30 m (raster)	NASA DEM
	Elevation	The mean sea-level altitude for each grid.	Reviewed every 5 years; Full update varies from 10-20 years		Polygon data	FEMA NFHL
	Flood hazard zone	A designated area classified by its level of flood risk based on probability, depth, and historical flooding patterns.				
	Tree canopy cover	The percentage of ground area covered by tree foliage when viewed from above.	Annual; last updated 2023	2023	30 m	USDA
	Vegetation distribution	Modified Soil Adjusted Vegetation Index (MSAVI) for vegetation detection.	16 days	Sept 2026 median composite	30 m	Landsat 9
	Reservoirs	Distance from reservoirs which include artificial and natural water bodies.	Biannual; last updated 2023	2023	Polygon data	FDEP
	Land cover	The physical material present on the surface; less detail	As needed; last updated 2025	2024		
Hazard	Precipitation	Total rainfall estimated for hurricanes.	30 minutes	Helene - Sept 25–28 2024 Milton - Oct 5–10 2024	~ 10 km (raster)	NASA IMERG
	Wind intensity	The 3-sec wind gust at 10 m elevation.	Event-based	Helene - Sept 27 2024 Milton - Oct 21 2024	~ 500 m (raster)	NIST/ARA
	Hurricane track	Distance from hurricane path to each grid.	Event-based	Helene- Sept 24 2024 Milton- Oct 5 2024	Line	National Hurricane Center
	Evacuation degree	The extent or level to which people have been relocated from a hazard-prone area.	Varies from as needed to annual	2023	Polygon data	FDEM

Flood extent	The spatial area covered by the flood water for each grid in percentage using SAR images.	Event-based	Helene - pre-event: 2024 9-14, 9-16, 9-19, 9-21, post-event: 9-26, 9-28, 10-3 Milton - pre-event: 2024 10-3, 10-8, post-event 10-11, 10-20	~ 20 m	NASA Disaster Mapping Portal
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325 **Table 4. Summary of the features description and their sources**

Model Framework

The proposed framework, named Debris-Cast, consists of a two-stage machine learning architecture designed to predict hurricane-generated debris volume at different grid levels. The framework integrates a debris occurrence classification model and an independent tiered regression model shown in Figure 7. The first stage identifies whether a grid cell contains debris, while the second stage estimates debris volume for debris-present observations. Two machine learning algorithms, RF and XGBoost, were implemented and evaluated within the same modeling framework.

The two-stage logic of separating debris presence from debris amount is not itself novel: González-Dueñas et al. (2023a) established this structure for hurricane debris. The contribution of Debris-Cast lies instead in four departures from that framework. First, its predictors are drawn jointly from the building, environmental, vegetation, and hazard domains using open geospatial sources, most of which have operational forecast analogues, rather than from a narrower feature set reported at the category level. Second, the second stage is a tiered regression that conditions volume estimation on debris magnitude, rather than a single global regression fitted to all debris-present cells. Third, the framework predicts multiple debris streams separately, whereas most prior hurricane debris models target a single aggregated debris quantity. Fourth, the spatial unit is treated as an experimental factor: H3 hexagonal and equivalent-area square grids are compared at two resolutions using an identical feature set and learning pipeline, whereas earlier gridded debris models fixed the unit to square grids. The novelty of this study therefore rests on the combination of multi-domain predictors, tiered regression, multiple debris categories, and spatial-grid comparison, rather than on the staged architecture itself.

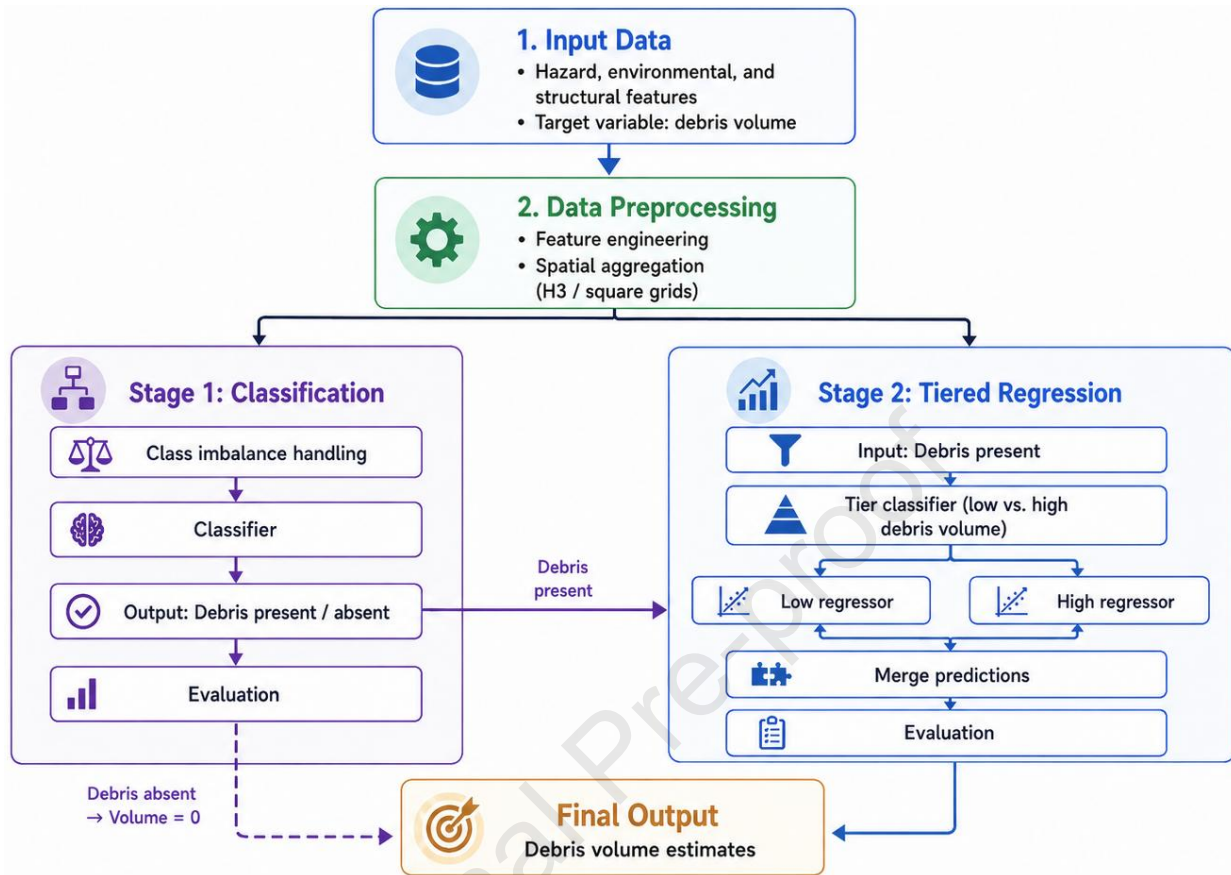


Figure 7. The two-stage machine-learning framework for debris estimation. Stage ML1 classifies whether each grid cell generates debris, and Stage 2 estimates debris volume for debris-present cells using a tiered low- and high-volume regression design, with Random Forest and XGBoost applied at each stage.

Data Preprocessing

The final dataset consisted of 21 predictor feature groups and 3 target features (C&D debris volume, vegetative debris volume and total debris volume). Hurricanes are a combination of multiple hazards such as wind, surge and rainfall, that interact with the built and natural environment to generate and transport debris (González-Dueñas et al., 2023a). To capture the sequential dual-hurricane exposure, for each spatial grid we used the maximum wind gust, the maximum rainfall, and the closest distance to the hurricane track across the two storms. After the features were collected and preprocessed, they were aggregated into four spatial grid units: H3 resolution 8 (H8), H3 resolution 9 (H9), and their approximately equal-area square grids (S8 and S9). Although H8 and S8, and H9 and S9, were designed to have approximately equivalent cell areas, they do not produce identical numbers of cells because the two tessellations differ in geometry, orientation, and alignment relative to the irregular study-area boundary. Study area boundary cells were further filtered by retaining only those with more than 90% of their original grid area, which resulted in slightly different final sample sizes between the corresponding hexagonal and square

grids. As described in the post-processing section above, grids within a 1 km buffer of temporary debris collection sites, grids with incomplete city records, and grids overlapping water bodies were also excluded. The final number of grid cells, or observations, retained for model development is presented in Table 5.

Table 5. Grid types and number of observations used in modeling. Original number of observations are number of grids clipped to Pinellas County boundary before post-processing, which consists of sea water and inland water bodies.

Model	Unit Area (sq km)	Original number of observations	Final number of observations
Hexagon grid Resolution 8 (H8)	~0.75	1,428	777
Hexagon grid Resolution 9 (H9)	~0.1	6,908	5,783
Square grid equal to H8 (S8)	~0.75	1,368	696
Square grid equal to H9 (S9)	~0.1	6,341	4,758

Numerical variables exhibiting high skewness were subjected to outlier clipping, in which extreme values exceeding data-driven percentile thresholds were excluded to reduce the influence of outliers. Selected variables associated with building characteristics, structural values, content values, and flood exposure were then transformed using a natural logarithm transformation to reduce skewness and stabilize variance. Distance-related variables were converted into binary indicators representing proximity to the coastline and reservoirs. Categorical variables without an inherent ranking, including building types, and occupancy types, were transformed using one-hot encoding (converting categorical data into binary numerical arrays). Variables with an ordinal structure, including evacuation degree and flood zone designation, were transformed using ordinal encoding to preserve their relative ordering.

Machine Learning Algorithms

RF is a versatile ensemble-based machine learning algorithm capable of performing both classification and regression tasks. The RF classifier constructs multiple unpruned decision trees, each trained on a bootstrap sample of the training dataset using a randomly selected subset of predictor variables (Breiman, 2001). The final classification is determined through a majority voting mechanism (hard voting), in which each decision tree casts a vote for a particular class and the class receiving the most votes is selected as the final prediction. In the regression stage, RF averages the outputs of the individual decision trees to produce a continuous value rather than a discrete class label. This dual capability makes RF particularly well suited to two-stage modeling frameworks, in which classification and regression are performed sequentially to first identify the presence of a phenomenon and then quantify its magnitude.

Extreme Gradient Boosting (XGBoost) is a highly efficient and scalable ensemble-based machine learning algorithm capable of performing both classification and regression tasks (Chen & Guestrin,

2016). XGBoost is based on gradient boosting, in which a sequence of weak learners (decision trees) is constructed iteratively, with each successive tree correcting the residual errors of its predecessor. For classification, XGBoost optimizes a log-loss objective to predict discrete class labels, whereas for regression it minimizes a mean-squared-error objective to generate continuous predictions. Unlike conventional boosting methods, XGBoost uses a regularized objective that penalizes model complexity, thereby reducing overfitting and improving generalization. Its computational efficiency, built-in handling of missing values, and strong predictive accuracy have made XGBoost a widely adopted algorithm for both classification and regression applications (Chen & Guestrin, 2016).

A. Classification Model

The first stage of the framework predicts the presence or absence of debris within each grid cell. For this purpose, the debris volume target was converted into a binary response grid cells with no debris were assigned a value of zero and grid cells with measurable debris were assigned a value of one. The dataset was randomly divided into 85% training and 15% testing subsets. This split was retained because the available debris dataset is limited and highly skewed, with relatively few high-volume debris observations. Spatial block partitioning would further reduce the effective training sample and could produce validation blocks containing few or no representative high-volume observations, resulting in unstable comparisons across folds. Therefore, the present random split evaluates model performance for unseen observations within the sampled spatial domain but does not provide a fully spatially independent estimate of geographic generalization.

A common issue in classification tasks is class imbalance in the raw data, in which a model overlooks the patterns of minority classes and produces results biased toward the majority classes. Several methods have been proposed to address class imbalance; among them, we applied the Synthetic Minority Oversampling Technique (SMOTE) to our dataset. SMOTE helps overcome the overfitting that can arise from random oversampling, which simply duplicates existing examples in the minority class (Chawla et al., 2002; Kim et al., 2022).

To reduce model complexity and improve computational efficiency, an initial feature importance analysis was conducted on the training data. Predictor variables were ranked by their relative contribution to model performance, and the most influential predictor groups were retained for subsequent model development. Both the RF and XGBoost classifiers were trained on balanced training datasets and used to predict whether debris was present within each grid cell. Hyperparameters were optimized using grid search on the training dataset.

The hyperparameters of the RF and XGBoost classifiers and regressors are summarized in Table 6.

Table 6. Model hyperparameters used in modeling

Model	Hyperparameters (value)
RF	Number of trees (n_estimators), Maximum depth of each tree (max_depth), Maximum number of features considered at each split (max_features), Minimum number of samples required at a leaf node (min_samples)
XGBoost	Number of boosting rounds / trees (n_estimators), Maximum depth of each tree (max_depth), Learning rate, Fraction of training samples used for each tree (subsample), Fraction of features used to build each tree (colsample_bytree)

Classification Evaluation Metrics

To evaluate the classification model, we used a combination of complementary metrics, including overall accuracy, confusion-matrix analysis, and the Receiver Operating Characteristic–Area Under the Curve (ROC-AUC). Accuracy is the ratio of correctly predicted observations to total observations; although simple and intuitive, it can be misleading when the dataset is imbalanced. The ROC curve visualizes the trade-off between the true positive rate (sensitivity) and the false positive rate ($1 - \text{specificity}$) across classification thresholds, while the AUC provides a single scalar summary of the model's overall discriminative ability (Fawcett, 2006). An AUC approaching 1.0 indicates strong discrimination, whereas a value of 0.5 indicates performance no better than random chance. Using multiple metrics together supports a thorough and reliable assessment of the classification model's performance.

B. Regression Model

The second component of the proposed framework is an independent tiered regression architecture developed to estimate debris volume. Unlike conventional single-model regression approaches, it consists of three components: a tier classifier and two specialized regression models for low-volume and high-volume debris. The regression architecture was trained exclusively on observations with non-zero debris volumes, ensuring that all models were calibrated on verified debris-present samples, with feature selection performed independently of the debris presence/absence classification model. A separate 85:15 train-test split was applied to the non-zero debris dataset. As in classification, an initial feature-importance analysis was conducted on the training data, and the most influential predictor groups were retained for further model development.

To account for heterogeneity in the debris-volume distribution, a systematic threshold-optimization procedure was implemented. Multiple configurations combining outlier-clip percentiles and volume-tier stratification thresholds were evaluated. Samples exceeding the clip percentile were excluded from training to reduce the adverse effects of extreme values. Within each filtered dataset, samples were partitioned into low-volume and high-volume groups based on the stratification threshold. Only configurations that maintained adequate sample sizes and

reasonable class balance (a 30–70% split) were retained for evaluation. To assess the relative influence of the predictor variables on debris generation, model-based feature importance was used for feature selection. Feature importance scores were obtained from the fitted Random Forest and XGBoost models and used to rank the predictors according to their contribution to model prediction. For categorical predictors represented by one-hot encoded variables, including building type, occupancy type, and landcover, the importance scores of the corresponding dummy variables were summed to obtain a single importance score for the original predictor. The predictors were then ranked based on their importance scores, and the 20 highest-ranked predictor groups were retained for subsequent model development.

Training and Inference

Following threshold selection, a tier classifier was trained to distinguish low-volume from high-volume observations. As in the first-stage classifier, SMOTE was applied to the training dataset to mitigate class imbalance between the two debris-volume groups. Both Random Forest and XGBoost classifiers were evaluated for tier classification.

Two independent regression models were then trained on their respective subsets: a low-volume regressor and a high-volume regressor. Before model fitting, debris-volume values were transformed using a natural-logarithm transformation to reduce skewness and stabilize variance. By training a separate model within each debris-volume tier, the framework lets each regressor specialize in a narrower range of debris quantities and capture tier-specific relationships that a single global regression model may not adequately represent.

During inference, the tier classifier first assigned each sample to a volume tier, after which the corresponding regressor generated the debris-volume estimate. The outputs of the two regressors were then combined to produce a unified prediction of debris volume across all debris-present grid cells. Log-transformed predictions were back-transformed to their original units, and all predictions were constrained to non-negative values. Hyperparameter tuning was performed separately for the low-volume and high-volume regression models, and the optimal parameter combinations were selected based on test R^2 .

Regression Evaluation Metrics

The predictive accuracy of the regression model was assessed using two widely adopted metrics: the coefficient of determination (R^2) and the mean absolute error (MAE) (He et al., 2026; Khoshvaght et al., 2025). R^2 quantifies the proportion of variance in the observed debris-volume values that the model accounts for, serving as an indicator of goodness of fit. An R^2 closer to 1.0 reflects stronger agreement between predicted and observed values, whereas a value approaching 0 indicates poor explanatory power. Because R^2 does not capture the actual magnitude of prediction errors in measurable units, MAE was used as a complementary metric. MAE is the average absolute difference between predicted and observed debris-volume values, expressed in the original unit of measurement (m^3), and provides a more tangible and interpretable measure of

prediction error. To enable comparison of prediction errors across grid resolutions with different spatial supports, MAE was normalized by grid-cell area as

$$\text{MAE}_{\text{area}} = \frac{\text{MAE}}{A},$$

where MAE_{area} represents the area-normalized mean absolute error (m^3/km^2), MAE is the raw per-cell mean absolute error (m^3), and A is the grid-cell area (km^2). Together, R^2 , MAE, and area-normalized MAE characterize overall model fit, cell-level prediction error, and prediction error relative to spatial support.

Results

We summarize the Stage 1 classification performance of the RF and XGBoost models across debris categories and spatial units (Table 7). In this stage, the models classify whether each grid cell contains debris or no debris. Overall, both machine learning algorithms achieved moderate to strong classification performance, with AUC values ranging from 76.55% to 93.90% and classification accuracies ranging from 68.57% to 89.36%.

For C&D debris, both RF and XGBoost produced consistently high Stage 1 classification performance across the evaluated spatial units. The highest performance was achieved by the RF model at H9 with an AUC of 93.90% and an accuracy of 89.17%, followed by XGBoost at H9 with an AUC of 92.52% and an accuracy of 89.36%. Across all spatial units, C&D debris generally showed the strongest classification performance among the three debris categories for both algorithms.

For vegetative debris, RF models produced AUC values averaging approximately 88.50% across all spatial units, while XGBoost models averaged 87.04%. The highest Stage 1 classification performance for vegetative debris was achieved by XGBoost using the S9 grid, with an AUC of 90.27% and an accuracy of 83.29%. For the combined debris category, Stage 1 classification performance varied more noticeably across spatial units. RF achieved its highest performance at H9 with an AUC of 90.12% and an accuracy of 82.14%. XGBoost produced comparable results at H9 with an AUC of 89.70% and an accuracy of 82.45%, and at S9 with an AUC of 89.57% and an accuracy of 83.19%. Compared with C&D and vegetative debris, the combined debris category showed greater variability across spatial units, particularly for the RF model.

Across the four spatial units, H9 consistently produced the highest or near-highest Stage 1 classification performance for both algorithms and most debris categories. The S9 spatial unit also produced competitive results, especially for vegetative and combined debris using XGBoost. In contrast, H8 generally produced slightly lower classification performance than H9, while S8 showed comparatively lower performance for RF in the combined debris category. These results indicate that finer spatial resolutions generally improved the model's ability to distinguish debris-present grids from debris-absent grids in Stage 1.

Table 7. Overall classification results (debris existence) of two ML algorithms for different grid-debris type combinations

Spatial unit	RF			XGBoost		
	AUC (Accuracy) in %			AUC (Accuracy) in %		
	CD	VG	Combined	CD	VG	Combined
H8	87.1 (79.49)	88.8 (79.49)	89.0 (76.92)	89.12 (82.05)	83.60 (80.12)	83.56 (80.76)
H9	93.9 (89.17)	89.53 (81.34)	90.12 (82.14)	92.52 (89.36)	89.99 (82.97)	89.70 (82.45)
S8	88.06 (82.86)	86.94 (79.05)	76.55 (68.57)	92.68 (89.28)	84.33 (82.14)	86.74 (84.28)
S9	90.6 (86.27)	88.73 (80.39)	87.63 (80.11)	90.45 (87.60)	90.27 (83.29)	89.57 (83.19)

Figure 8 presents the top 20 feature groups for the best-performing classification models for three debris types (RF–H9 for C&D and combined debris, and XGBoost–S9 for vegetative debris). Across all debris types, hazard-related variables, particularly the nearest distance to the hurricane track and maximum wind gust, consistently emerged as the most influential predictors as expected. Beyond hazard intensity, building age and construction material were among the primary drivers of C&D debris generation, which is intuitive as buildings are the source of construction and demolition. In contrast, landcover, occupancy type and building types were top three major contributors of vegetative debris followed by hazard intensities. In our dataset, occupancy types consisted of residential (92.7%), industrial (1.6%), commercial (5%) and other categories (0.5%). Residential buildings dominate and are accompanied by lawns, gardens, trees etc. and when these vegetative sources interact with hazard intensity produce vegetative debris. Hazard intensities and building types contributed more prominently to combined debris prediction. Flood-related variables, including post-flood extent and flood zone, also contributed to model performance but were consistently less influential than direct hurricane hazard characteristics.

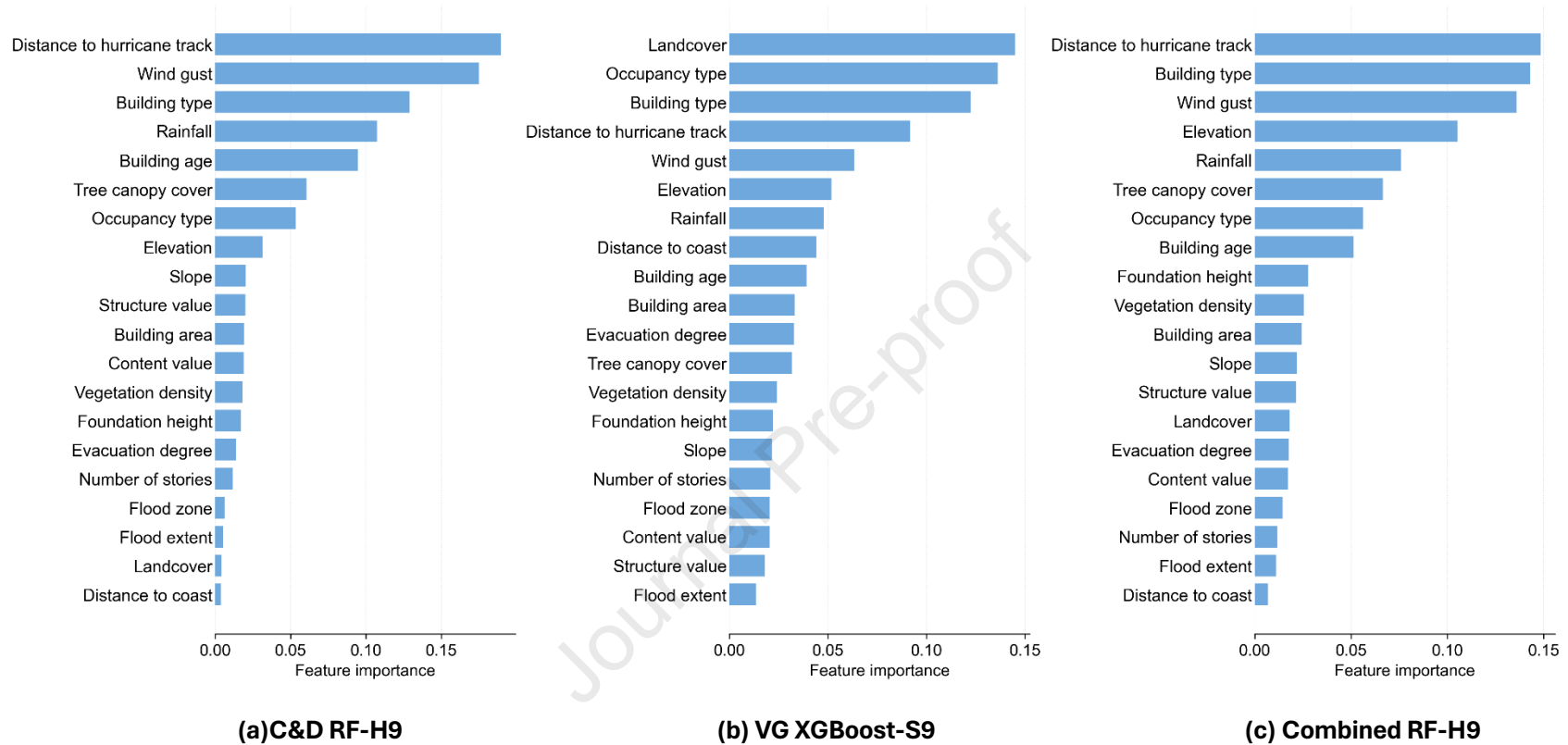


Figure 8. Top 20 most important feature groups for the best-performing classification model of each target variable (Random Forest at H9 for C&D and combined debris, and XGBoost at S9 for vegetative debris), showing the dominance of hazard-related predictors such as distance to the hurricane track and maximum wind gust.

After debris-present grid cells were identified in Stage 1, the Stage 2 models estimated debris volume using independent tiered regression models. Table 8 reports the regression performance across debris categories, spatial units, and machine learning algorithms.

Table 8 summarizes the Stage 2 regression performance across debris types, spatial units, and machine-learning algorithms. Model performance varied by debris type, grid configuration, and algorithm. For C&D debris, the highest R^2 was obtained by XGBoost using S8 ($R^2 = 0.846$), followed by XGBoost using H8 ($R^2 = 0.842$) and RF using H8 ($R^2 = 0.824$). For vegetative debris, RF with H8 achieved the highest overall R^2 (0.869), while XGBoost with H8 produced a comparable R^2 of 0.843. For combined debris, XGBoost with S8 achieved the highest R^2 (0.845), followed closely by RF with H8 ($R^2 = 0.839$) and RF with S8 ($R^2 = 0.823$).

Raw per-cell MAE generally decreased at the finer H9 and S9 resolutions because each fine-resolution cell represents a substantially smaller spatial area and contains less debris volume. For example, RF C&D MAE decreased from 83.69 m^3 at H8 to 20.85 m^3 at H9, while RF vegetative MAE decreased from 511.11 to 81.92 m^3 . Similarly, XGBoost C&D MAE decreased from 435.95 m^3 at H8 to 179.78 m^3 at H9. These raw MAE values describe prediction error at the individual grid-cell level but should not be interpreted as directly comparable measures of accuracy across resolutions with different cell areas.

To provide a common spatial basis for comparison, MAE was also normalized by grid-cell area (m^3/km^2). The area-normalized results show that the lower raw MAE at finer resolutions did not consistently correspond to lower prediction error per unit area. For RF, C&D area-normalized MAE increased from 111.59 m^3/km^2 at H8 to 208.50 m^3/km^2 at H9, and vegetative debris increased from 681.48 to 819.20 m^3/km^2 . Combined debris similarly increased from 769.61 m^3/km^2 at H8 to 1,096.30 m^3/km^2 at H9. For the square grids, RF C&D area-normalized MAE decreased from 358.75 m^3/km^2 at S8 to 264.00 m^3/km^2 at S9, whereas vegetative debris increased from 580.76 to 871.90 m^3/km^2 and combined debris increased substantially from 223.24 to 1,518.40 m^3/km^2 .

A similar pattern was observed for XGBoost. Area-normalized MAE increased from H8 to H9 for all three debris categories: from 581.27 to 1,797.80 m^3/km^2 for C&D, from 762.56 to 1,283.10 m^3/km^2 for vegetative debris, and from 1,267.13 to 2,678.80 m^3/km^2 for combined debris. For the square grids, area-normalized MAE increased from S8 to S9 from 477.11 to 1,179.50 m^3/km^2 for C&D, from 857.80 to 1,554.70 m^3/km^2 for vegetative debris, and from 1,006.77 to 1,961.70 m^3/km^2 for combined debris.

These results indicate a scale-dependent tradeoff rather than a uniform improvement in predictive accuracy with finer spatial resolution. Fine-resolution grids produced lower absolute prediction errors per cell in many cases and provide greater spatial detail for identifying localized debris patterns. However, after accounting for differences in cell area, coarse-resolution grids generally produced lower error per unit area, particularly for vegetative and combined debris. The results therefore suggest that grid resolution affects both the magnitude and spatial interpretation of

594 prediction error, and that the preferred resolution depends on whether the application prioritizes
595 localized spatial detail or lower error over a larger spatial support.

596

597 **Table 8. Stage 2 regression results for debris volume prediction across debris types, spatial units, and**
 598 **machine learning algorithms**

Spatial unit	RF			XGBoost		
	R^2			R^2		
	(MAE m^3)			(MAE m^3)		
	[MAE _{area} m^3/km^2]			[MAE _{area} m^3/km^2]		
	CD	VG	Combined	CD	VG	Combined
H8	0.824 (83.69) [111.59]	0.869 (511.11) [681.48]	0.839 (577.21) [769.61]	0.842 (435.95) [581.27]	0.843 (571.92) [762.56]	0.743 (950.35) [1267.13]
H9	0.763 (20.85) [208.50]	0.787 (81.92) [819.20]	0.777 (109.63) [1096.30]	0.797 (179.78) [1797.8]	0.812 (128.31) [1283.1]	0.730 (267.88) [2678.8]
S8	0.801 (269.06) [358.75]	0.798 (435.57) [580.76]	0.823 (167.43) [223.24]	0.846 (357.83) [477.11]	0.770 (643.35) [857.80]	0.845 (755.08) [1006.77]
S9	0.761 (26.40) [264.00]	0.794 (87.19) [871.90]	0.761 (151.84) [1518.40]	0.830 (117.95) [1179.5]	0.774 (155.47) [1554.70]	0.784 (196.17) [1961.70]

599

Figure 9 compares the distributions of observed and predicted combined debris-removal volumes for the H9 and S9 test cells. The purpose of this figure is to evaluate whether each model reproduces the observed frequency, spread, and range of debris volumes, rather than to assess cell-by-cell correspondence.

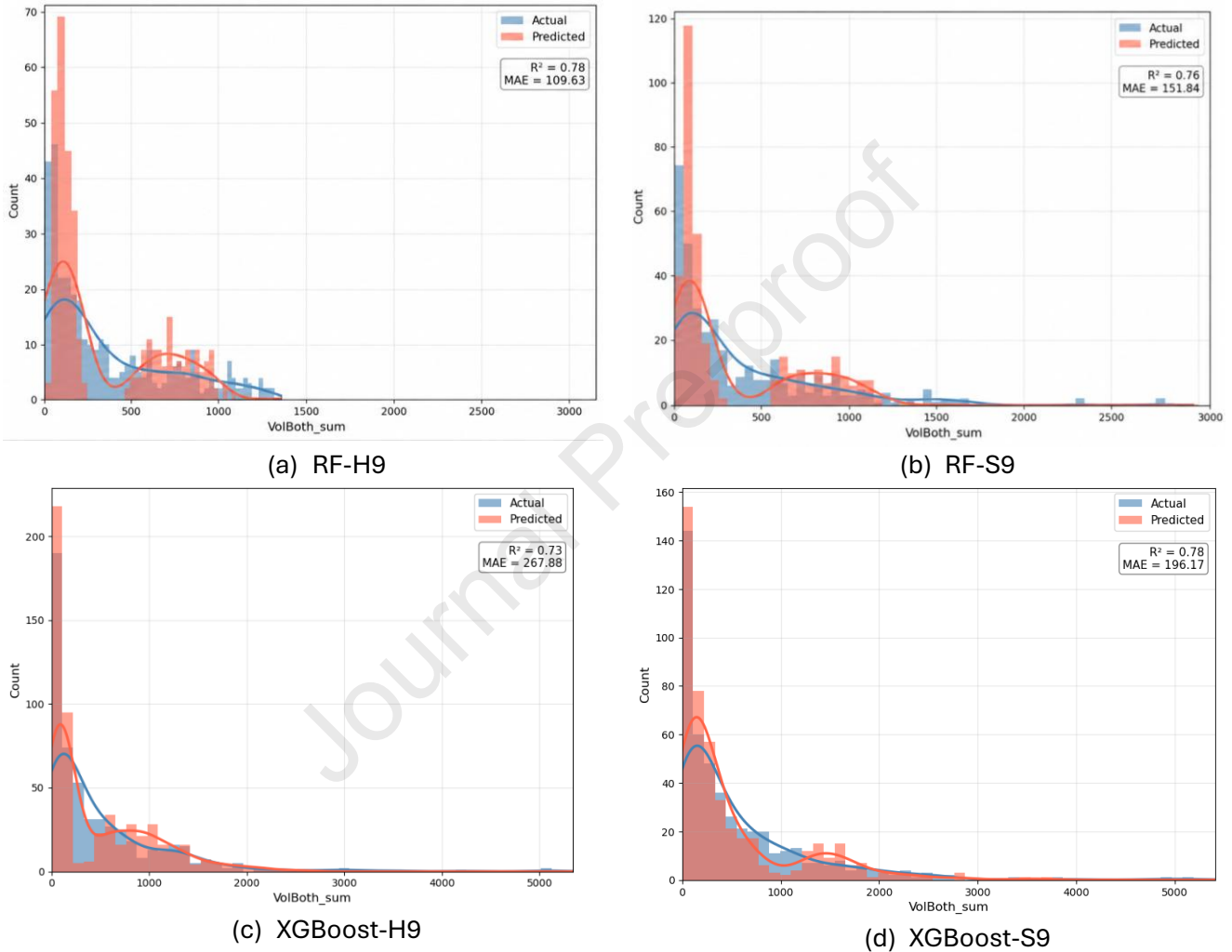


Figure 9. Distributions of observed and predicted combined debris-removal volumes across the H9 and S9 test cells for the Random Forest and XGBoost models. The histograms and density curves compare the frequency, spread, and upper tail of observed and predicted volumes.

Figure 10 visualizes the final Stage 2 debris volume prediction outputs for the H9 test dataset, comparing ground-truth combined debris volumes with predicted volumes from the RF and XGBoost models. The maps show that both models captured the general spatial distribution of debris volume, with higher debris concentrations mainly observed in the southern portion of Pinellas County. However, the RF prediction more closely follows the ground-truth pattern, which is

613 consistent with its lower MAE. The zoomed panels further show that both models reproduce local
614 debris clusters, but cell-level differences remain, especially for higher-volume grids.

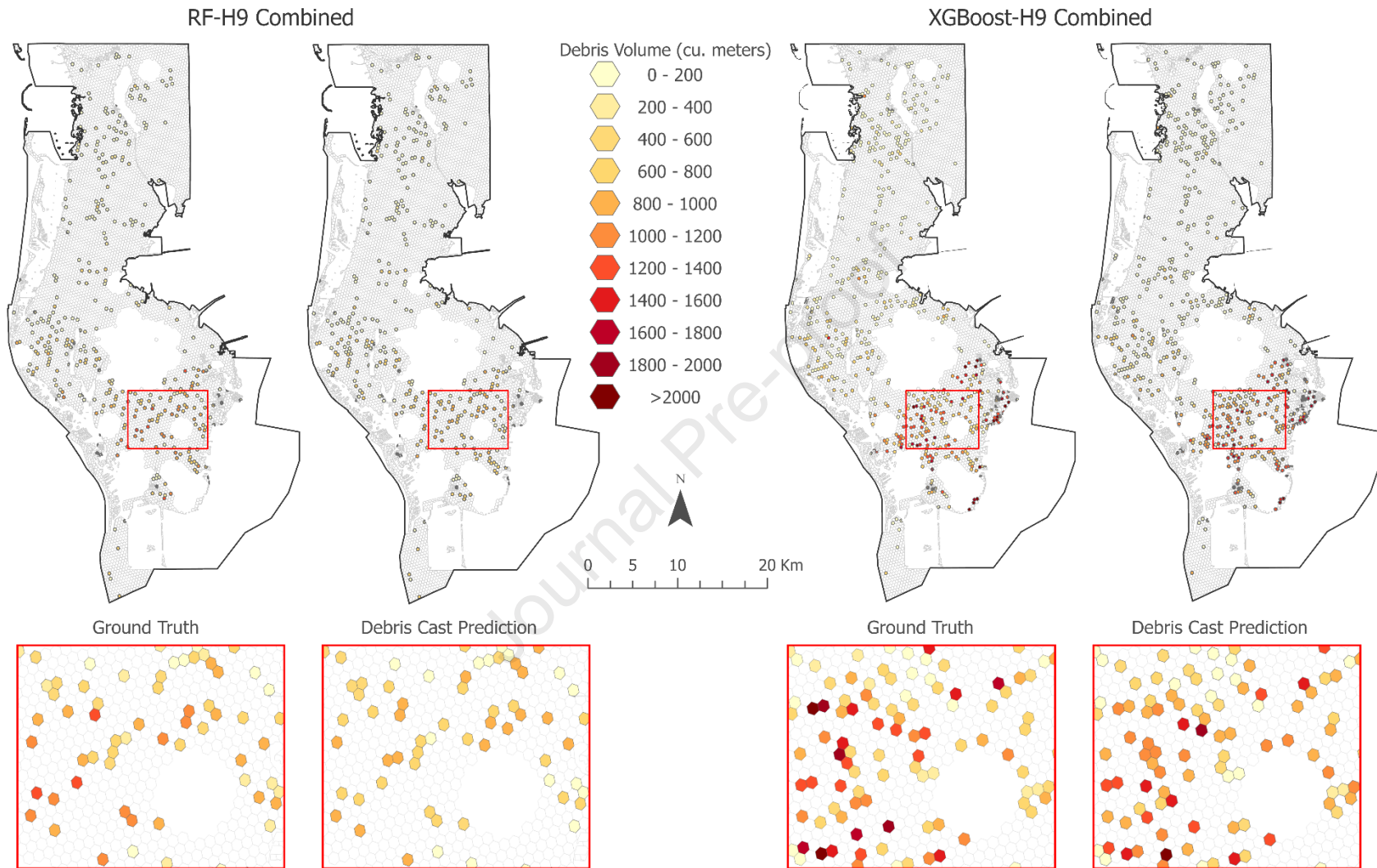


Figure 10. Spatial comparison of ground-truth and predicted combined debris volumes for RF and XGBoost models using the H9 test dataset. The two ground-truth maps represent different model-specific subsets of the observed data. Red boxes indicate the areas enlarged below

Discussion

The main contribution of this study is the demonstration that hurricane debris estimation can be reframed as a rapid, spatially explicit learning problem in which debris occurrence, debris magnitude, and spatial support are modeled jointly rather than treated as separate operational afterthoughts. The results show that multi-source geospatial predictors contain strong information for identifying debris-generating locations, with classification performance reaching an AUC of 93.9% for construction and demolition debris at H3 resolution 9.

The regression results further reveal a scale-dependent tradeoff. Coarser grids generally showed stronger overall model fit and lower area-normalized prediction error, while finer grids provided greater spatial detail and lower raw per-cell MAE. Because raw MAE is influenced by grid-cell area, the lower values observed for H9/S9 should not be interpreted as direct evidence of greater predictive accuracy. For example, H8/S8 cells cover approximately 0.75 km², whereas H9/S9 cells cover approximately 0.10 km². After accounting for these differences in spatial support, area-normalized MAE was generally lower for the coarser grids, particularly for vegetative and combined debris. These results suggest that the preferred spatial resolution depends on the intended application. Coarser grids may be more suitable for regional-scale estimation where broader spatial patterns and lower error per unit area are prioritized, whereas finer grids can provide more localized debris information that may be useful for neighborhood-level screening, routing, and resource planning. Thus, the operational value of finer grids lies primarily in their greater spatial detail rather than uniformly greater predictive accuracy.

Importantly, hexagonal and square grids performed comparably across debris types and resolutions, indicating that H3 can be adopted for operational integration without an obvious loss of predictive performance. These findings strengthen the paper's central argument that machine learning can capture nonlinear interactions among hazard, built-environment, vegetation, and terrain variables, while also showing that spatial-index design is an important component of deployable debris intelligence. The practical implication is that H3 should be framed not as universally more accurate than square grids, but as a viable spatial framework whose value lies in standardization, multi-resolution integration, and compatibility with hazard decision-support systems.

Overall, RF and XGBoost exhibited comparable classification performance in Stage 1 across the evaluated spatial units. Of the twelve model configurations, RF achieved higher AUC values in six cases and XGBoost achieved higher observed performance than RF in the remaining six. Although both algorithms showed similar overall predictive capability, XGBoost produced a narrower range of AUC values (83.56–92.68), indicating more consistent performance across debris categories and spatial units, whereas RF showed a wider range of AUC values (76.55–93.90), yielding both the highest performing and the lowest-performing model among the evaluated configurations. The similar performance of both algorithms suggests that the selected hazard, environmental, and building variables provide strong discriminatory information for identifying debris occurrence

regardless of the underlying ensemble learning technique. Nevertheless, the slight advantage of XGBoost in some datasets likely reflects its sequential boosting strategy, which iteratively corrects errors from previous trees and is generally more effective at capturing complex nonlinear relationships and interactions among predictors. The bagging approach of RF, by contrast, tends to be more robust to noise and overfitting but may be less effective when the decision boundary is highly heterogeneous (Imani et al., 2025; Li et al., 2022). This difference is evident for the S8 combined debris dataset, where XGBoost achieved higher observed performance than RF, suggesting that the combined debris class involves more complex feature interactions that boosting captures more readily. Despite these differences, both algorithms consistently achieved high classification performance across debris categories, indicating that the proposed framework is insensitive to the choice of ensemble classifier.

For regression, RF and XGBoost showed distinct performance characteristics across all spatial units and debris categories. Although XGBoost achieved higher R^2 in several cases, especially for C&D and combined debris, these gains were accompanied by larger prediction errors. RF, by contrast, generally produced a more balanced trade-off between explanatory power and predictive accuracy, particularly in the finer-resolution grids (H9 and S9). The higher R^2 values occasionally achieved by XGBoost are likely attributable to its boosting strategy, which sequentially minimizes residual errors while capturing complex nonlinear relationships and interactions among predictors. Previous studies have noted that XGBoost can build stronger learners but may also exhibit greater variance and increased sensitivity to the characteristics of the training data, particularly in relatively small or heterogeneous datasets (Li et al., 2025). Consequently, this added flexibility can improve explained variance while producing larger absolute prediction errors when applied to highly skewed or heterogeneous debris distributions. The ensemble-averaging mechanism of RF, on the other hand, tends to produce more conservative predictions, reducing variance and improving overall prediction robustness. The discrepancy between R^2 and MAE observed across several experiments further underscores the importance of evaluating multiple performance metrics when assessing disaster-impact estimation models, since higher explained variance does not necessarily translate into smaller prediction errors. Considering both metrics together, the RF models at H9 and S9 showed the most balanced and practically useful predictive performance across the debris categories.

The results showed that both spatial resolution and grid geometry influence debris prediction performance, although their influence differs between debris-occurrence classification and volume estimation. For both RF and XGBoost, classification performance improved at higher-resolution grids, with H9 and S9 consistently producing higher AUC and accuracy values than coarser grids. These findings indicate that high-resolution spatial units better preserve the localized hazard impacts and other features that distinguish debris-generating from non-debris-generating locations. Finer grid cells reduce the spatial averaging of hazard intensity, land cover, building characteristics, and exposure conditions, allowing the machine learning algorithms to identify patterns associated with debris occurrence more effectively.

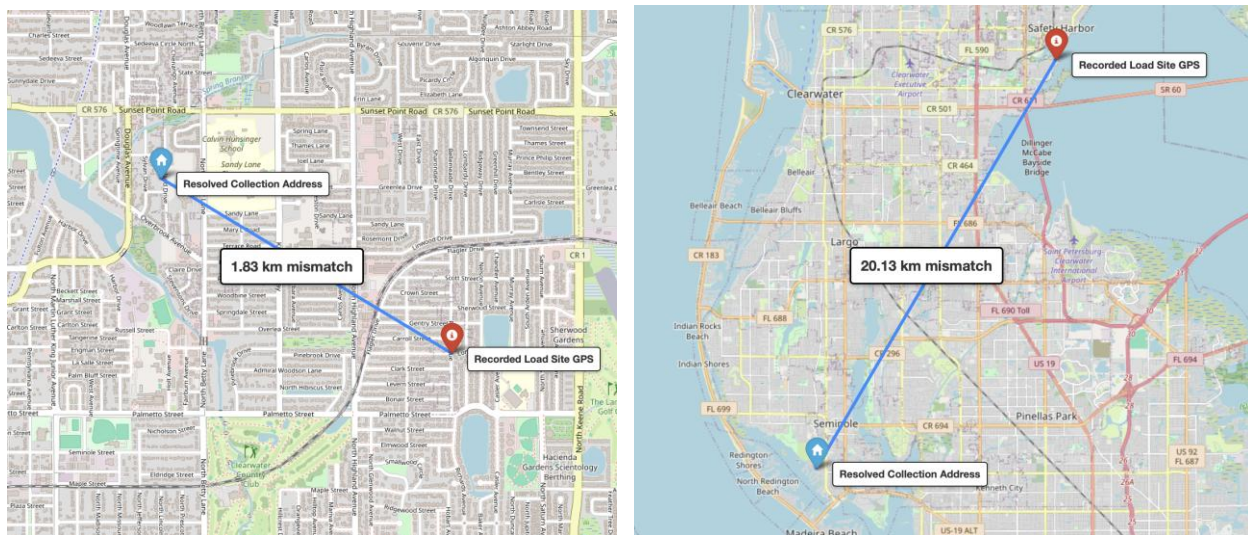
By contrast, the regression results reveal a trade-off between explanatory power and predictive accuracy across spatial resolutions. Coarser spatial units (H8 and S8) generally produced higher R^2 values, suggesting that aggregation at larger spatial scales reduces local variability and allows models to explain broad spatial trends more effectively. Finer spatial units (H9 and S9), conversely, consistently yielded lower MAEs, indicating improved prediction accuracy at the local scale. This pattern likely reflects the ability of higher-resolution grids to better capture localized heterogeneity in hurricane impacts, land cover, and built-environment characteristics, thereby reducing absolute prediction error despite explaining a smaller proportion of overall variance. The trade-off underscores the importance of selecting spatial units according to the intended application, since models optimized for explanatory performance may not provide the most accurate predictions for operational decision-making.

Furthermore, these findings can be interpreted through the lens of the MAUP, which recognizes that analytical results may vary with the size and configuration of the spatial units (Hayward, 2009; Pollack et al., 2022). Changes in grid resolution alter the degree of spatial aggregation and may affect the classification and regression tasks differently. The sensitivity of model performance to both grid size and geometry therefore highlights the importance of carefully selecting the unit of analysis when developing spatially explicit debris-prediction models.

Despite the advantages of hexagonal grids on preserving curvilinear and radial spatial patterns and more accurately characterizing neighborhood relationships, our results indicate that hexagonal grids did not universally outperform the square grids across all model configurations. Performance was generally comparable between the two grid geometries, with the relative advantage varying by debris type, model, and resolution. For classification, both geometries achieved high predictive performance at finer resolutions, suggesting that spatial resolution had a stronger influence than cell shape on the ability to detect debris occurrence. Similarly, the regression results showed no consistent dominance of either geometry, indicating that both tessellations could capture the underlying relationships among hazard exposure, built-environment characteristics, and debris generation. The absence of a clear predictive advantage may indicate that machine learning algorithms are flexible enough to learn these complex relationships regardless of grid geometry, particularly when sufficient predictor variables are available. Even without a clear predictive advantage, hexagonal grids offer conceptual and methodological strengths for disaster-related analysis and decision-making, particularly in representing spatial continuity and the complex propagation of hazard impacts. Thus, although grid geometry may not substantially change predictive outcomes, it remains a meaningful design choice when spatial interpretation or process-based modeling is a priority. These advantages have already led to the adoption of hexagonal indexing systems in operational platforms such as the Texas Disaster Information System (TDIS) (accessed through <https://portal.cloud.tdis.io/>), which employs the H3 spatial indexing framework to support hazard visualization, situational awareness, and disaster planning for Texas communities.

The debris datasets consist of load tickets generated during debris removal operations. Each record contains two location identifiers: the address where debris was collected and the GPS coordinates logged at the collection site. Comparing the two revealed meaningful inconsistencies across the dataset. About 14% of records had a collection address and GPS coordinate separated by more than 100 meters. Of those mismatched records, 46% were associated with temporary debris staging sites or parks, which explains the large spatial offset. Excluding those records, roughly 8% of the full dataset still contained location mismatches exceeding 100 meters. For grid-based modeling, minor offsets can be absorbed through spatial aggregation, but extreme mismatches like the example in Figure 11 introduce noise that can degrade model performance, particularly at finer spatial resolutions where a misplaced record lands in the wrong grid cell entirely. Additional uncertainty arose from errors in the reported collection addresses. Missing cardinal directions and incorrect street type suffixes made some addresses difficult to geolocate with confidence, limiting our ability to verify record locations against GPS coordinates.

Future work will explore validation and resampling strategies specifically designed for spatially structured and highly skewed debris data, including spatial blocking, grouped resampling, and independent event- or region-level holdouts as additional datasets become available. The spatial inconsistencies found in the load-ticket records point to a deeper structural limitation in how debris data are currently collected and reported. Address-GPS mismatches affected roughly 14% of records overall, and after removing cases tied to staging sites and parks, 8% of the full dataset still showed offsets exceeding 100 meters (see Figure 11). These discrepancies are not simply noise; they reflect a recording system designed primarily for contractor reimbursement rather than spatial analysis. At coarser resolutions, grid-level aggregation can absorb minor positional errors, but at finer resolutions such as H9 and S9, a misplaced record can land in a completely different grid cell, directly corrupting the training labels the model depends on. This helps explain, in part, why finer-resolution models may face a higher noise floor that limits how much classification and regression performance can improve regardless of the predictor set or algorithm used.



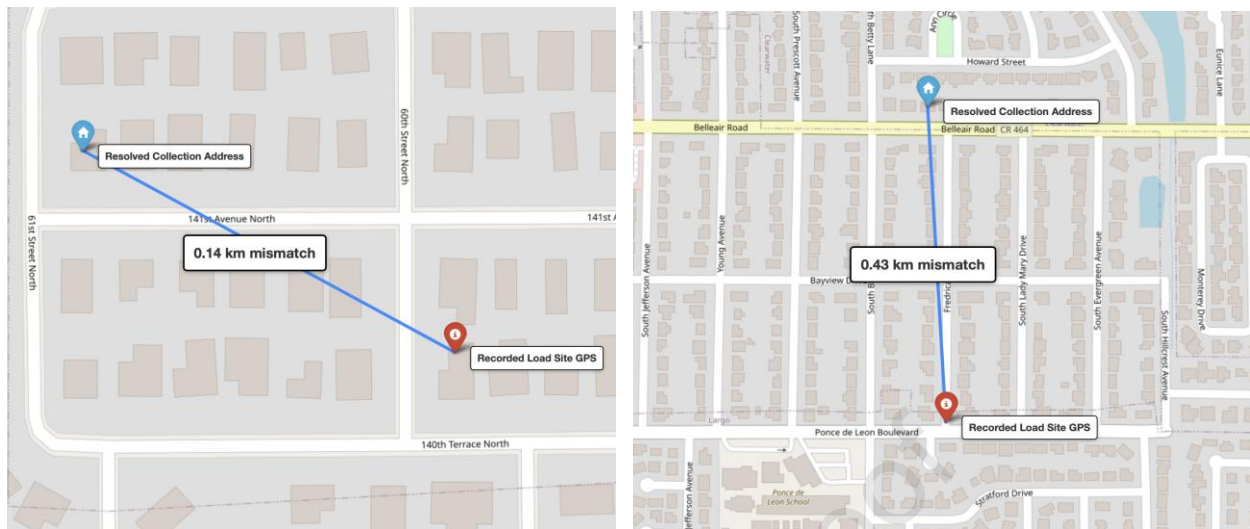


Figure 11. Examples of spatial mismatch between actual debris collection locations and the coordinates recorded in load-ticket records, illustrating the address-versus-GPS discrepancies that introduce locational noise into model training, particularly at finer spatial resolutions.

For each debris-hauling trip, one load ticket is initiated at the curbside pickup location by the load site monitor and completed at the temporary debris site or final disposal site by the disposal site monitor. The same ticket documents the pickup location, truck information, debris type, disposal location, estimated quantity, and monitor signatures. A separate ticket is created only when debris is reloaded from the temporary debris site and hauled to another destination, such as a landfill, recycler, or final disposal facility. Load-ticket records were not originally designed for debris volume estimation, and the current schema reflects that. Because a ticket is generated for each trip, the same physical debris can appear in the dataset more than once (see Figure 5). Debris picked up from a curbside and hauled to a temporary debris management site generates one ticket. When that same debris is later moved from the staging site to a final disposal facility, a second ticket is generated for the same material. If both records are included in a debris estimation dataset without distinguishing the trip type, the total volume is inflated. In this study, that problem was identified only through spatial visualization, which revealed dense clusters of collection points at known staging locations. Those clusters were flagged, buffered, and excluded from the analysis to prevent double counting. This manual step worked here but cannot be automated or reliably replicated without explicit trip-type information in the raw data. A straightforward addition to the load-ticket schema would be a required field that identifies whether each trip originated from a curbside collection or a staging site transfer. That single change would make the dataset directly usable for debris volume estimation without requiring spatial review, extending the value of a system that already captures rich operational information.

These observations suggest that future debris data collection efforts could benefit from more intentional dataset design. Several considerations worth exploring include:

- Extending ticketing procedures to track debris through every transfer point to final disposal, not just the first pickup, would establish a more complete chain of custody; a future revision of the FEMA-325 guidance could make this linkage explicit
- Clearer documentation at the time of public data release, including notes on what the dataset does and does not capture, would help researchers interpret records more accurately. In this study, for example, 32,854 St. Petersburg records had no debris type recorded, representing over one million cubic meters of unclassified volume (see Table 3: Disaster debris dataset breakdown by debris type)
- Working toward more consistent data schemes across counties and municipalities would make cross-event and regional comparisons more feasible

Improvements in these areas would likely reduce noise in the debris training data and support more reliable model performance, particularly at finer spatial resolutions where data quality constraints are most consequential.

In addition, comparisons among RF and XGBoost and among the spatial grid configurations are based on point estimates of AUC, R^2 , MAE, and area-normalized MAE. Formal statistical significance tests were not conducted; therefore, differences among model and grid configurations should be interpreted as descriptive rather than statistically established. Future work should incorporate confidence intervals and paired statistical comparisons to quantify uncertainty in these performance differences.

Conclusion

This study developed a two-stage machine learning framework to predict hurricane-generated debris volume across four spatial grid configurations using multiple open geospatial datasets. The first stage identified whether debris was present within each grid cell, and the second stage estimated debris volume only for cells where debris was detected, using both RF and XGBoost. The framework was applied to debris records from Hurricanes Helene and Milton in Pinellas County, Florida, across three debris types: construction and demolition, vegetative, and combined.

Classification performance improved at finer grid resolutions regardless of grid geometry, with the highest AUC reaching 93.9% for C&D debris using Random Forest at H9. This confirms that finer spatial units better preserve the localized hazard and built-environment patterns that distinguish debris-generating from non-debris-generating locations. Regression performance followed the opposite trend. Coarser grids produced higher R^2 values, the best reaching 0.869 for vegetative debris at H8, consistent with the Modifiable Areal Unit Problem. Finer grids, however, yielded lower prediction errors, with MAE as low as 20.85 m³ for C&D debris at H9, offering a more balanced trade-off between model fit and practical accuracy. These findings highlight that relying on a single metric such as R^2 is insufficient; both explained variance and prediction error must be considered together when evaluating debris-prediction models.

Hexagonal and square grids produced comparable predictive performance across all debris types and resolutions, indicating that machine learning algorithms can capture debris-environment relationships regardless of cell shape. Even so, H3 hexagonal grids offer practical advantages for operational use. The H3 framework is already adopted across geospatial platforms and hazard-modeling tools, so debris estimates produced in H3 can be integrated with other hazard datasets and decision-support systems (e.g. locating point of distribution (PODs), shelter, evacuation plan) without spatial reprocessing. Hexagonal grids also better represent curvilinear hazard patterns such as wind swaths and surge propagation, which adds value when the framework is used alongside process-based models. For emergency managers, this means debris hot spot maps generated by this framework can be directly overlaid with storm surge forecasts and wind fields inside existing dashboards, eliminating an extra spatial conversion step. Public works departments can use the same outputs to identify staging site locations, and debris contractors can receive prioritized deployment zones tied to high-volume grid cells. State recovery agencies can draw on the volume estimates for early cost projections before cleanup contracts are finalized.

A practical implementation pathway should proceed in four phases. In the pre-event phase, local and state emergency-management agencies, public works departments, and debris contractors would assemble the static geospatial predictor base such as building inventories, elevation, land cover, tree canopy, flood zones, evacuation zones, and water-proximity features within a standardized H3 or equivalent grid framework. In the event-monitoring phase, forecast wind, rainfall, surge, and flood layers would be ingested as they become available, converted to the same spatial grid, and used to generate preliminary debris-occurrence probabilities and volume estimates at multiple lead times. In the response phase, model outputs would be reviewed through a human-in-the-loop process with emergency managers and debris coordinators, who would validate hot spots, adjust for local knowledge, identify staging-site needs, and prioritize contractor deployment. In the recovery-learning phase, observed load tickets, field reports, and disposal records would be reconciled with predicted outputs to recalibrate the model, update error distributions, and improve transferability for future events. Key barriers include forecast uncertainty, inconsistent debris records, data-sharing constraints, and municipal variation in cleanup workflows; these can be mitigated through uncertainty bands, standardized ticketing fields, documented preprocessing rules, and routine post-event model audits. Implemented in this way, the framework becomes not a replacement for expert judgment, but a rapid decision-support layer for anticipatory debris planning and early recovery coordination.

In its current form, the framework is best positioned as a rapid screening tool that can prioritize areas for field validation and resource planning. The next step toward full pre-landfall deployment is substituting the observed hazard fields used here with true operational forecast products for wind, rainfall, surge, and flood extent, and validating how debris estimates change across forecast lead times. That validation step will determine how far in advance reliable estimates can be generated and which debris types and grid resolutions are most sensitive to forecast uncertainty. Until then, the framework provides a practical and immediately usable improvement over fixed empirical tools,

one that emergency managers can deploy with open geospatial data and standard spatial indexing infrastructure already available in most state and local agencies.

Several limitations define the scope of this study and point to clear future research needs. First, the empirical evaluation is based on a single county affected by two sequential hurricanes, which limits generalizability across regions, building stocks, debris-management practices, and single-event scenarios. In addition, the current evaluation relies on a random train-test split, which may produce optimistic performance estimates because neighboring grid cells can share similar hazard and built-environment characteristics. Spatial block cross-validation would provide a stricter assessment of geographic generalization, but its application to the present dataset is challenging because of the limited number of observations and the highly skewed debris-volume distribution. Future work should therefore evaluate spatially grouped validation, event-level holdouts, and municipality-level holdouts using larger and more geographically diverse datasets. Application to another municipality or future event may also require local retraining or calibration, and direct transfer of the fitted model has not yet been demonstrated. Second, the current framework is evaluated using historical observed and model-derived hazard fields rather than true pre-landfall forecast inputs. Although several hazard variables have operational forecast analogues, the SAR-derived flood extent used in this study has no direct pre-landfall equivalent and would need to be replaced by forecast inundation guidance or omitted. Therefore, the present results demonstrate retrospective and screening capability rather than validated real-time forecasting or operational deployment. Future studies should evaluate lead-time-specific wind, rainfall, surge, and flood forecasts and quantify how prediction performance changes under forecast uncertainty. Third, operational load-ticket records are imperfect labels because they contain missing debris-type fields, address/GPS mismatches, staging-site effects, and incomplete municipal coverage. These inconsistencies can introduce noise into both classification and regression targets, particularly at finer spatial resolutions. Future work should develop standardized debris-data schemas, improved chain-of-custody tracking, sensitivity analyses for excluded records, and uncertainty-aware label construction. Finally, comparisons among RF and XGBoost and among spatial grid configurations are based primarily on point estimates of AUC, R^2 , MAE, and area-normalized MAE. Formal statistical significance tests were not conducted; therefore, observed differences should be interpreted as descriptive rather than statistically established. The present study addresses differences in spatial support by reporting area-normalized MAE in addition to raw MAE, but future work should further strengthen model and grid comparisons using confidence intervals, paired statistical tests, and aggregation of fine-resolution predictions to a common spatial support.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: